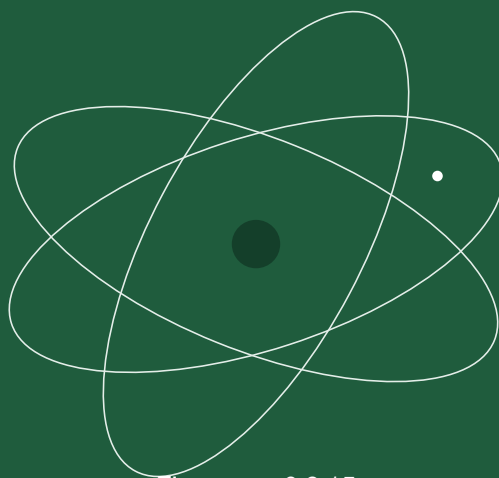


CardSat

The Complete Operating Manual



Firmware v0.9.15
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A complete guide to operating **CardSat**, the amateur-radio satellite tracker and multi-radio CAT Doppler controller for the M5Stack Cardputer ADV (Icom, Yaesu, Kenwood).

Status. CardSat runs on the Cardputer ADV, and every feature has been exercised on hardware **except radio (CAT) control and the antenna rotator** — those two are still unverified on real equipment. Their math (the per-protocol CAT frequency encoders, and all four rotator backends – GS-232 framing, the rotctld TCP client, PstRotator UDP and the direct-Yaesu I2C interface) is host-tested, but nothing has driven an actual radio or rotator yet. Keep the serial monitor open and verify on the air before trusting it for a contact.

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01

Overview

CardSat turns a pocket-sized Cardputer ADV into a standalone satellite station controller. It:

- downloads GP (General Perturbations / OMM) orbital elements from AMSAT and transponder frequencies from the SatNOGS database, and caches both to flash for **fully offline** use;
- predicts passes with **SGP4** and shows live az/el/range, a polar sky plot, and per-pass elevation curves;
- drives an **Icom, Yaesu, or Kenwood** transceiver over **CAT** with continuous **Doppler correction**, keeping a constant frequency *at the satellite* (the AMSAT “One True Rule”) so your signal stays put in a linear transponder’s passband;
- lets you tune the passband from the device **or from the radio’s own knob**;
- maintains a **favorites** list, a unified **Next Passes** schedule, an **AOS alarm**, a **deep-sleep-until-next-pass** power saver, and **sun/eclipse** status;
- points an **az/el antenna rotator** – a wired **Yaesu GS-232** controller or a networked **Hamlib rotctld** server;
- **logs QSOs** on the device and **exports ADIF** for LoTW / eQSL or your main logger.

Everything runs on the device. WiFi is needed only to refresh GP/transponders and to set the clock.

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What you need

- **M5Stack Cardputer ADV** (ESP32-S3, 8 MB flash, 240×135 screen, 56-key keyboard, speaker, microSD, Grove port, 2×7 header).
 - **A supported transceiver** in one of three CAT families:
 - **Icom CI-V** — IC-820, IC-821, IC-910, IC-970, IC-9100, IC-9700;
 - **Yaesu** — FT-847, FT-736R;
 - **Kenwood** — TS-790, TS-2000.
 - **A CAT interface suited to that radio.** The ESP32-S3 pins are **not** 5 V tolerant — never connect CAT directly. **Icom:** a 3.3 V-safe CI-V interface (one-transistor circuit or a ready-made board). **Kenwood:** a **MAX3232** RS-232 level shifter on the DB-9 COM port. **Yaesu:** a serial CAT interface (verify TTL vs RS-232 per the CAT manual).
 - (Optional) a **GPS:** a NMEA receiver on the Grove port, or an **M5Stack Cap LoRa** (868 or 1262) module with onboard GNSS.
 - (Optional) **WiFi** to fetch GP/transponders and sync the clock.
-

03

Connecting your radio

All three CAT families share **UART1, RX = G1 / TX = G2** by default, but the **interface hardware differs** (see §2 and §16).

1. Wire the appropriate CAT interface between the radio's control jack (Icom **REMOTE**, Kenwood **COM DB-9**, Yaesu **CAT**) and **G1/G2**. **Never connect the radio directly to the GPIO pins** — the ESP32-S3 is not 5 V tolerant.
2. Note the radio's CAT settings: **baud** (all families) and, for **Icom**, the **CI-V address** (fixed on older rigs).
3. In CardSat, open **Settings**, choose your **radio model** (this auto-fills the defaults), then adjust **CAT baud** — and, for Icom, the **CI-V address** — to match.

CardSat drives two independent VFOs. By default **downlink = Sub/RX, uplink = Main/TX**, but the **VFO Type** setting can swap the roles (*Main Dn/Sub Up*). On **Icom** it manages MAIN/SUB and, by default, leaves the rig's satellite mode **off**; the **Sat mode** setting commands it on/off when you engage radio control (a no-op on rigs without one, such as the IC-820/821/910/970). On **Yaesu and Kenwood** the rig's own satellite / full-duplex mode and the band pair are set up **by you on the radio** — CAT can't switch bands on those rigs — and CardSat Doppler-tunes within them. (See §16.)

Icom over the network (no CI-V wiring)

The **IC-9700** can be controlled over **WiFi/Ethernet** instead of the wired CI-V bus — CardSat speaks the same RS-BA1 UDP protocol as Icom's own remote software. No level shifter, no UART: the radio's CI-V/G1/G2 pins stay free.

Scope. Icom LAN in CardSat is intended and tested only for the **IC-9700** — the one network-capable Icom in CardSat's radio list that is a full-duplex satellite rig. Other networked Icoms (IC-705, IC-7610, IC-785x) speak the same RS-BA1 protocol, so the transport would work, but they are single-receiver HF/VHF radios without the MAIN/SUB satellite architecture CardSat drives, so they are not supported here. Select **Icom LAN** only with the **IC-9700** chosen as the radio model.

On the radio (IC-9700: **MENU > SET > Network**): set **Network Control = ON**, set a **Network User1** id + password, leave the **Control** port at **50001** (Serial = 50002, Audio = 50003 follow it), and turn **CI-V Transceive ON** so the radio reports changes. Give the radio a stable IP (DHCP reservation or static).

In CardSat **Settings**: set **CAT type → Icom LAN**, **LAN host** to the radio's IP, **LAN port** to 50001 (unless you changed it), and **LAN user / LAN pass** to the Network User1 credentials. CardSat connects in the background; once the link is up the radio behaves exactly like a wired IC-9700 (MAIN/SUB, Doppler, sat mode, CTCSS all work the same). Only **CAT** is carried — CardSat does not open the audio stream. Keep the credentials on a trusted LAN. (See §16.)

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Connecting a GPS (optional)

GPS runs on **UART2**, independent of CAT. On the **Location** screen press **s** to cycle the source:

Source	Pins	Baud	Notes
Grove 9600	G1/G2	9600	! same pins as CAT — don't use both at once
Grove 115200	G1/G2	115200	! same pins as CAT — don't use both at once
Cap LoRa868	G15/G13	115200	M5Stack Cap LoRa (868) onboard AT6668 GNSS
Cap LoRa1262	G15/G13	115200	M5Stack Cap LoRa (1262) onboard AT6668 GNSS

Both Cap LoRa modules use identical GPS settings (115200 8N1 on G15/G13). For a Grove-port receiver, pick the baud that matches it.

Press **p** on the Location screen to enable/disable GPS use. With a fix, your latitude, longitude, altitude, and grid update automatically, and the clock is set from GPS time. The Location screen also refreshes on its own the moment a fix is gained or lost or the satellite count changes.

If you use the **Grove** GPS option you cannot use CAT on G1/G2 at the same time. The two **Cap LoRa** options use G15/G13 and coexist with CAT.

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Installing the firmware

There are two ways to get CardSat onto the device: flash a **prebuilt binary** (no toolchain needed — easiest), or **build from source** with the Arduino IDE or PlatformIO. Two binaries are published with each release:

File	Use it with	Notes
<code>CardSat.bin</code>	Launcher (bmorcelli/Launcher)	App-only image; Launcher writes the partition table and bootloader, so this file is only usable through Launcher — it cannot be flashed on its own.
<code>CardSat_Merged.bin</code>	M5Burner , or a direct flash (esptool / web flasher)	Complete standalone image (bootloader + partition table + app + empty filesystem), written at offset <code>0x0</code> .

Install with Launcher (`CardSat.bin`)

[Launcher](#) by bmorcelli is a firmware launcher for the Cardputer (and many other ESP32 boards) that installs and runs binaries from a microSD card, over-the-air, or through its WebUI — and builds the right partition layout for each app automatically. It must already be installed on the device (see its repository for how to flash Launcher itself).

1. Copy `CardSat.bin` to the Launcher binaries folder on the device's microSD card (or push it via Launcher's WebUI / OTA favorites).
2. On the device, start **Launcher**, browse to **CardSat**, and install/run it. Launcher writes the appropriate partition scheme and the app for you.
3. `CardSat.bin` is an app-only image with no standalone bootloader or partition table, so it works **only** through Launcher. To flash without Launcher, use the merged image below.

Install with M5Burner, or direct flash (`CardSat_Merged.bin`)

`CardSat_Merged.bin` is a **complete image** — bootloader, partition table, application, and the empty LittleFS filesystem combined, written at offset `0x0`. Use it with **M5Burner** (add it as a custom firmware and burn), for a fresh device, to recover one in an unknown state, or whenever you are not using Launcher.

To flash it directly with **esptool** (replace the port with yours):

```
esptool.py --chip esp32s3 --port /dev/ttyACM0 --baud 921600 \
  write_flash 0x0 CardSat_Merged.bin
```


Or with the web **ESP Tool** (<https://espressif.github.io/esptool-js/>): connect the Cardputer over USB, set chip to **ESP32-S3**, add **CardSat_Merged.bin** at address **0x0**, and flash.

Hold the device's reset/boot as your tool instructs if it does not enter download mode automatically. After flashing, the device reboots into CardSat. (The merged image starts with an **empty** filesystem; run **Update** once on first boot to download GP elements — see §7.)

Build from source — Arduino IDE (single file)

1. Install libraries: **M5Cardputer**, **ArduinoJson** (v7), **TinyGPSPlus** from the Library Manager, and the Hopperpop **Sgp4** library via *Sketch* → *Include Library* → *Add .ZIP Library* (from <https://github.com/Hopperpop/Sgp4-Library>).
2. Open **CardSat.ino**. Under **Tools**, set:
 - **Board:** ESP32S3 Dev Module (or M5StampS3)
 - **Flash Size:** 8MB
 - **Partition Scheme:** Huge APP (3MB No OTA/1MB SPIFFS) — *required*
 - **PSRAM:** Disabled · **USB CDC On Boot:** Enabled
3. Upload. Open the Serial Monitor at **115200** baud for diagnostics.

Build from source — PlatformIO

`pio run` to build, `pio run -t upload` to flash, `pio device monitor` to watch the log. The `cardputer_adv` env pins the libraries and the 8 MB partition layout.

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The keyboard: how to navigate

CardSat uses the arrow legends printed on the Cardputer keys:

Key	Action
;	up
.	down
,	left
/	right
ENTER	select / confirm
` or DEL	back / cancel
{ }	page up / page down (lists)
b	save a screenshot to the SD card (see §18)

Other letter keys are screen-specific actions and are shown in the **footer** at the bottom of each screen. When in doubt, read the footer. The one global exception is **b**, which saves a screenshot on any screen and is deliberately not shown in any footer.

07

First-time setup

1. **Settings** (;/. to move, ,// to change a value, ENTER to edit text):
 - **Radio** — select your model; CAT baud (and, for Icom, the CI-V address) auto-fill.
 - **CAT baud** — set to match the radio's CAT menu (applies to every CAT family).
 - **CI-V addr** — set to match the radio; **Icom only**.
 - **Min pass el** — passes whose **peak elevation** never reaches this value are hidden from the pass lists and schedule (default 5°).
 - **WiFi SSID / WiFi pass** — enter your network, then **Save & test WiFi**. On the **WiFi SSID** row you can instead press **s** to **scan** for nearby networks, pick one from the list (strongest first; * = secured), and then enter its password (open networks skip the password step).
 - **AOS alarm** — on/off for the pre-pass beeps.
 - **VFO Type** — which physical VFO carries each leg: *Main Up/Sub Dn* (default) or *Main Dn/Sub Up*.
 - **Sat mode** — command the rig's own satellite mode on/off when you engage CAT (a no-op on rigs without one).
 - **CAT rate** — how often Doppler/CAT updates are sent to the radio (default **500 ms**, adjustable in 10 ms steps). A soft floor keeps the *effective* rate no faster than the CAT baud can service one update; the row shows (**min N**) when your setting is being clamped.
 - **CAT delay** — how long to pause *after each CAT command* before sending the next one to the radio (default **70 ms**, 0–200 ms in 2 ms steps). Raise it if an older or slow radio drops or mis-handles back-to-back commands; lower it toward 0 for the fastest tuning on a radio that keeps up. CI-V (Icom) only at present.
2. **Location** — set your position one of three ways:
 - **e** latitude, **o** longitude, **a** altitude; or
 - **g** Maidenhead grid (uppercased as you type — hold shift for lowercase sub-squares — as on the log and mutual-pass grid fields); or
 - **p** enable GPS (and **s** to pick the GPS source).
 - If you have no network or GPS, press **c** to set the **UTC clock** manually (**YYYY-MM-DD HH:MM:SS**).
3. **Update** — press **k** (or ENTER) to download GP data; this also syncs the clock over NTP if WiFi is configured. Optionally press **a** to cache **all** transponders for offline use.
4. You're ready: **Satellites** to pick birds, **Next Passes** to see what's up.

At every power-on, if a WiFi network is configured CardSat connects automatically and sets the clock over **NTP** before loading data — so a network-connected unit boots with the correct time and no key presses. If WiFi is unavailable it carries on with GPS or the cached/manual clock; the attempt is best-effort and non-fatal (it can add a few seconds to boot while it tries). If the clock is set and even the freshest cached element set is **over a week old**, CardSat also refreshes GP automatically at boot. The display blanks its backlight after the **Screen sleep** idle time (never while tracking or alarming); any key wakes it.

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Screen reference

CardSat is a simple state machine. From **Home** you reach every screen; ``` or **DEL** always steps back.

Home

A menu: **Satellites** · **Next Passes (all favs)** · **Passes (sel)** · **Track (sel)** · **Location** · **Update** · **Settings** · **About / diagnostics** · **Log**. The currently selected satellite is shown at the bottom right. `;/.` move, **ENTER** selects.

About

Author credit (**Paul Stoetzer, N8HM**), firmware version and build date, storage backend (microSD or internal flash), GP catalog size and freshest element age, WiFi/IP, battery level, free heap, and uptime. ``` or **ENTER** returns home.

Logging QSOs (Log)

There are two ways to start an entry:

- **From a pass** — press **l** on the **Track** or **Polar** screen to log the contact you're working; radio control keeps running while you type. CardSat snapshots the UTC time, satellite, mode and the live **Doppler-corrected** uplink/downlink frequencies, plus your grid and callsign (from **My callsign** in Settings).
- **After the fact** — choose **New QSO entry** from the **Log** menu when you're *not* working a pass. The entry opens with the time set to now and everything ready to edit, so you can log a contact you made earlier or on another rig.

Every field is editable on the entry screen — `;/.` move, **ENTER** edits the selected field:

- **Date / Time** — UTC, entered as **YYYY-MM-DD** and **HH:MM:SS**.
- **Sat** — **ENTER** opens the satellite list; pick a bird and CardSat fills the **Sat**, **Mode** and frequency fields with its defaults — the **non-Doppler centre** of a linear transponder's passband, or the **nominal** downlink/uplink for an FM or single-channel satellite.
- **DL MHz / UL MHz** — downlink and uplink in MHz (e.g. **145.960**).
- **Call** — the station worked (required to save). Letters default to **uppercase** (hold shift for lowercase).
- **Mode** — on a **linear** transponder, **ENTER** toggles **SSB** ↔ **CW** for this contact; on an FM bird the mode is fixed.
- **RST S / RST R** — report sent and received (default **59**).
- **Grid** — the other station's grid, **uppercase** by default (optional, but grids are the point of sat ops).
- **Notes** — free text.

s saves, ``` cancels. QSOs are appended to **/CardSat/qso_log.csv** (one row each; **notes** is the last column, so commas there are fine).

The **Log** item on the main menu offers **New QSO entry**, **View / edit log**, and **Export to ADIF** (writes `/CardSat/qso_log.adi`, including `STATION_CALLSIGN` from My callsign, for upload to LoTW/eQSL or import into your main logger).

LoTW limits the `SAT_NAME` field to six characters and uses its own names, so on export CardSat translates each satellite via `/CardSat/lotw_sats.csv` (rows of `SAT_NAME`, `AMSAT_NAME`; a built-in table covers the common cases if the file is absent). If a logged satellite has no match, CardSat prompts you for its `SAT_NAME` (entry capped at six characters, uppercased) and saves it back to the CSV, so it's only asked once. Update that CSV on the card as new satellites appear.

View / edit log is a scrollable list of recent contacts (`;/.` to move). Open one with **ENTER** to correct **any field** — including the **date, time, satellite and frequencies** — then **s** to save, or press **x** twice to delete it; changes are written straight back to the CSV. The most recent **120** QSOs are available on the device — the complete log always lives in `/CardSat/qso_log.csv`, which you can also read or edit on a computer.

Satellites

The catalog (up to 220 sats from the GP data, plus any you add manually).

- `;/.` move, `{/}` jump 10 rows.
- **f** — toggle **favorite** (favorites are marked with *****).
- **v** — toggle **favorites-only** filter.
- **n** — add a **manual GP satellite**: enter the name, NORAD ID, epoch, then each orbital element in turn (see **Edit** under §8).
- **x** — **delete** the selected satellite, but *only if it was added manually* (one you entered with **n**). Press **x** once to arm, **x** again to confirm. This removes it from `/CardSat/mgp.json` and from your favorites; cached GP satellites from the network can't be deleted this way (they'd just return on the next Update). To **edit** a manual satellite, delete it and re-add it with **n**.
- **e** — open the **EQX table** (equatorial crossings) for the selected satellite, for use with an OSCARLOCATOR (see **Transponder/EQX screens** in §8).
- **ENTER** — load the satellite's transponders and open its **Passes**.
- **`** — back to Home.

At the right edge each row may show an **AMSAT activity mark**: a **filled dot** means the satellite has been reported **heard** (active) recently, a **filled square** means it has only been reported **telemetry only** (a beacon/telemetry is alive but no transponder or voice contact was logged), a **hollow ring** means it has only been reported **not heard** recently, and **no mark** means there are no recent reports. When a satellite has a mix of reports the strongest wins (heard > telemetry > not heard). The marks come from the [AMSAT OSCAR Status page](#) and are refreshed whenever you run **Update** (see §GP source). Matching is by base designator, so it works for AMSAT-named birds; satellites loaded from CelesTrak categories usually won't match and simply show no mark.

Orbital analysis (o)

Opened with **o** from **Satellites**. Nine pages, paged with `,//`, **r** to recompute, **`** to leave.

8.0.5.1 Theory of operation — how these numbers are produced A short primer, because the pages below assume some of this:

The elements. Each satellite is described by a set of **mean orbital elements** (the modern GP/OMM form of the classic two-line element set, TLE). The six that fix the orbit's size, shape and orientation are: **semi-major axis** a (orbit size, here derived from mean motion), **eccentricity** e (how elliptical, 0 = circle), **inclination** i (tilt of the orbit plane to the equator), **right ascension of the ascending node** Ω /RAAN (where the orbit plane cuts the equator going north, measured around the equator), **argument of perigee** ω (where the low point sits within the orbit plane), and **mean anomaly** M (where the satellite is along the orbit at the element epoch). **Mean motion** n (revolutions/day) stands in for a , and **B*** carries atmospheric drag. "Mean" means small periodic wobbles have been averaged out; the propagator puts them back.

The propagator. CardSat predicts position with **SGP4** — the same Simplified General Perturbations model the elements were built for. SGP4 is not a plain Kepler solver: it folds in the dominant perturbations (Earth's oblateness through the **J2** harmonic, plus atmospheric drag via B*) so that a near-Earth satellite's real motion is reproduced to roughly a kilometre near epoch. Every live read-out — az/el, range, range-rate, sub-point — comes from propagating the elements to the current instant and converting the resulting position/velocity vectors. **Accuracy decays with element age**, which is why the Info and Orbit-position pages show how old the set is and CardSat nudges you to refresh GP. For the full theory of GP/OMM elements, what SGP4 is and isn't, the coordinate frames involved, and where prediction error comes from, see §14.

Range rate and Doppler. The radial velocity (range-rate) is taken straight from the SGP4 velocity vector — the component along the station→satellite line. Doppler shift is then $\Delta f = -(\text{range-rate} \div c) \cdot f$: positive range-rate (receding) lowers the received frequency, and the shift scales with frequency, so a 70 cm downlink swings roughly three times as far as a 2 m one. This is the same quantity the Track screen uses to tune the radio (see §9).

J2 and why pass times drift. A spherical Earth would hold the orbit plane fixed in space; the real Earth's equatorial bulge (the J2 term) makes the plane **precess**. The two secular rates CardSat reports come directly from J2: the node regresses at $\dot{\Omega} = -1.5 n J2 (Re/p)^2 \cos i$ (°/day) and perigee rotates at $\dot{\omega} = 0.75 n J2 (Re/p)^2 (5\cos^2 i - 1)$ (°/day), where $p = a(1-e^2)$. Node regression is what walks a Sun-relative orbit's pass times earlier each day, and when it happens to equal the Earth's ~0.986°/day motion around the Sun the orbit is **sun-synchronous** (it keeps a fixed local solar time — the **LTAN**). Apsidal precession turns ω , moving where perigee sits; it goes to zero at the **critical inclination** of 63.4° (where $5\cos^2 i - 1 = 0$), the trick Molniya orbits use to keep perigee parked.

Beta angle and eclipses. The **solar beta angle** β is the angle between the Sun and the orbit *plane* (CardSat computes it as 90° minus the angle between the orbit-normal vector and the Sun vector). It governs lighting: at low β the satellite dips into Earth's shadow once per revolution; past a threshold $\beta^* = \arccos(Re/(Re+h))$ the orbit rides above the shadow and is in **continuous sunlight**. CardSat tests shadow with a **cylindrical-umbra** model (Earth casts a parallel-sided shadow tube, no penumbra) — simple, and good to about a minute for power-budget purposes. As the node precesses (above), β cycles over weeks, producing **eclipse seasons** and full-sun seasons; the Sun/Beta and Illumination screens visualise this.

Footprint geometry. The ground footprint is the cap of Earth from which the satellite is above the horizon. Its diameter is $2 \cdot Re \cdot \arccos(Re/(Re+h))$, set purely by altitude h — higher means a wider circle and a longer maximum possible contact. For an elliptical orbit the footprint breathes between its apogee (largest) and perigee (smallest) values, both of which the Info page lists.

The pages:

- **Info** — NORAD, current altitude, **footprint** diameter, period, apogee/ perigee altitudes, the **footprint diameters at apogee and perigee**, inclination/eccentricity, semi-major axis, B^* with a rough drag-decay estimate, element age/revolution, and the next ascending-node longitude/time. The decay estimate integrates B^* through an exponential-atmosphere model with a **King-Hele** treatment that lets an eccentric orbit circularize (drag at perigee lowers apogee fastest), down to a ~120 km reentry. The physics: drag acts hardest where the air is densest, which is at **perigee**, and a velocity loss there pulls down the *opposite* side of the orbit — so an elliptical orbit rounds off (apogee drops toward perigee) before the whole thing spirals in. B^* scales how much atmosphere the satellite “feels” (ballistic coefficient $\times$ a reference density), and the model decrements energy each revolution until the perigee reaches the ~120 km reentry floor. Because thermospheric density swings about a factor of ten over the solar cycle - the single biggest driver of lifetime - a second “**Decay rng**” line shows the bracket from **solar maximum (shortest)** to **solar minimum (longest)**, with the assumed level (**mean/min/max/auto**, set in *Settings -> Station / display -> Decay solar*) in parentheses. In **auto** the density scale is derived from the live **F10.7 solar flux**, downloaded with the GP elements and cached (the setting row shows the current value); without a cached flux it falls back to mean. Treat all of this as **order-of-magnitude**: it ignores attitude, lift, and short-term space weather, and B^* itself is often a fitted fudge term, so the chart is a “should I worry about this object” cue, not a reentry prediction. The **footprint** is the diameter of the visibility circle on the ground ($2 \cdot R_e \cdot \cos(R_e / (R_e + h))$) — the widest separation between two stations that can both see the bird at once, i.e. the **longest theoretically possible QSO** through it. It grows with altitude, so for an elliptical orbit the apogee figure is the best case and perigee the worst.
- **Live** — az/el, range, range-rate, Doppler at 145.8/435 MHz, sub-point, mean anomaly/phase, the sunlit/eclipse flag, and the **eclipse depth** — the PREDICT-style angle (degrees) by which the satellite is inside Earth’s umbral shadow (positive when eclipsed, negative as a sunlight margin), so you can see not just *whether* it’s in shadow but *how deeply*. The depth is the angular gap between the satellite’s anti-solar position and the edge of the shadow cylinder: it climbs as the bird moves toward the centre of the shadow and crosses zero exactly at the **terminator** (entry/exit), which is why a value near 0° means a sunrise/sunset is imminent — useful for spin-stabilised or solar-powered birds whose behaviour changes at the shadow boundary.
- **Next pass** — AOS countdown, duration, max elevation, AOS/LOS azimuths, sunlit fraction, eclipse entry/exit, the **peak eclipse depth** if the bird transits shadow during the pass, the optical-visibility flag, the **slant range at AOS/TCA/LOS** and the **one-way path delay** at closest approach ($\text{range} \div c$).
- **Ground track** — the next two orbits projected on an equirectangular map.
- **Doppler** — the Doppler curve across the next pass, computed at a **user-settable beacon frequency** (press **f** to change it; default 145.8 MHz). Below the plot it shows the peak Doppler shift and the maximum range-rate of the pass.
- **Nodal** — orbit-plane dynamics from the J2 secular model: revolutions per day; **node drift** (RAAN regression, $^\circ/\text{day}$) and **perigee drift** (apsidal precession, $^\circ/\text{day}$); a **sun-synchronous** flag (node drift $\approx +0.986^\circ/\text{day}$); the **LTAN** (local solar time of the ascending node); an approximate **repeat ground-track** cycle (revs/days, ≤ 30 days); and the **longest possible pass** (an overhead pass at apogee). The repeat cycle looks for a small whole number of days in which the satellite completes a whole number of revolutions,

so its track over the ground closes back on itself — when it exists, the same passes recur on that period (a 3-day repeat means today’s geometry returns in three days). These are first-order estimates — handy for understanding why pass times march earlier each day, whether a bird is sun-synchronous, and how long the very best passes can run — not a precision propagator.

- **Sun / Beta** — the orbit’s lighting geometry. The **solar beta angle** (the angle between the Sun and the orbit plane) now, the analytic β^* threshold beyond which the orbit is in continuous sunlight, and whether the bird is currently in a **full-sun orbit** or **eclipsed every revolution** — the verdict and the **eclipse fraction / minutes per orbit** come from sampling a full orbit with the same geometric shadow test the Illumination screen uses, so the two never disagree. A final line scans ahead to the **next transition** (the date the orbit next crosses into full sun, or back into eclipse, with a day countdown). Useful for solar-powered birds and for anticipating eclipse-season power behaviour.
- **Pass outlook** — a planning summary over the next 7 days: the total number of passes above your mask, how many clear 30°, the longest pass, and the mean gap between passes — plus the **best upcoming pass** (its peak elevation, the date/time it occurs, the countdown to it, and its duration). This answers “when this week is worth operating this bird” at a glance.
- **Orbit position** — where the satellite is in its orbit right now: mean anomaly, true anomaly, and **argument of latitude**, the **time to the next perigee and apogee**, plus argument of perigee, RAAN, the current revolution number, and the element-set age. The three angles describe position three ways: **mean anomaly** advances at a constant rate (it’s the orbit clock, uniform in time); **true anomaly** is the real geometric angle from perigee (it runs ahead near perigee where the bird moves fast, behind near apogee — the two are equal only for a circular orbit); and **argument of latitude** ($\omega + \text{true anomaly}$) is the angle up from the equator, which tells you how far through the north/south part of the orbit it is. For eccentric orbits the perigee/apogee timing shows when the bird is moving slowest and sitting highest — i.e. when the longest-dwell passes occur.

Next Passes (schedule)

One merged list of the next pass for **every favorite**, soonest first. Each row shows the countdown to AOS (or **NOW** if the bird is already up), the satellite name, maximum elevation, and pass length. A red **!** flags a satellite whose elements are stale (see §14).

- **;/.** move.
- **ENTER** — jump straight to **Track** for that satellite.
- **m** — open the live **World map** (all footprints; see **World map**).
- **r** — recompute the schedule.
- **z** — **deep-sleep** until ~60 s before the next AOS (see §12).
- **`** — back.

Passes

Upcoming passes for the selected satellite, in UTC: **AOS time**, **duration**, **max elevation**, **LOS time**. The element-set age is shown top-right, color-graded.

- **;/.** — select a pass.
- **d** — open the **pass-detail plot** (below).
- **t** or **ENTER** — go to **Track**.

- **n** — add a **manual transponder** for this satellite (see below).
- **r** — recompute the pass list.
- **x** — **mutual window**: enter a remote grid to find co-visibility passes (below).
- **v** — **10-day pass overview** (InstantTrack-style visibility chart; below).
- **i** — **illumination** (60-day Sun/eclipse raster; below).
- **`** — back to Satellites.

Pass detail

An **elevation-vs-time curve** for the selected pass. The curve is drawn **yellow where the satellite is sunlit** and **blue where it is in eclipse**. A cyan vertical line marks “now” if the pass is in progress. Below the plot: AOS time and azimuth, LOS time and azimuth, maximum elevation, pass duration, and the percentage of the pass spent in sunlight. Press **p** for a **polar view of this pass** — its ground track across the sky with A/L (AOS/LOS) markers and an arrow showing the direction of travel; **p** toggles back to the curve. **`** or ENTER returns to Passes.

Mutual windows (co-visibility)

From **Passes**, press **x** and enter a remote station’s **Maidenhead grid** (4 or 6 characters, e.g. **I091** or **JN58td**). CardSat scans the selected satellite’s upcoming passes **over the next 10 days** and lists every window where **both you and that station have the satellite above your horizons at the same time** — the periods you could actually make a contact through the bird. Each row shows the **start time (UTC)**, the **duration (m:ss)**, and the **maximum elevation at each end** — yours (**me**) and theirs (**dx**) — so you can judge how workable the window is. **;/.** scroll; **`** returns to Passes.

Co-visibility is computed to the **0° horizon** for both stations; a window with very low max elevations may be hard to use in practice. The remote grid is assumed at sea level.

10-day pass overview (v)

From **Passes**, press **v** for an at-a-glance chart of the selected satellite’s passes over the **next 10 days**, modelled on InstantTrack’s “Multiple Days for Single Satellite” visibility screen. Each row is one day (today at the top), drawn as a 24-hour timeline (00–24 h UTC, left to right) with faint gridlines at 06/12/18 h. The window is aligned to UTC midnight and spans ten **full** days, so every row is filled edge to edge — the chart is not cut off partway through the last day at the time you opened it. Every pass is a coloured bar from AOS to LOS, shaded by peak elevation — **dim green** below 15°, **green** to 40°, **yellow** above — so high passes stand out. A red tick on the top row marks the current time. **;/.** **scroll one day at a time** — the oldest day falls off the top and a new day appears at the bottom (you can scroll forward indefinitely, but not before today). **r** recomputes; **`** returns to Passes.

Illumination (i)

From **Passes**, press **i** for a **60-day solar-illumination raster** in the style of DK3WN’s *illum*. The horizontal axis is date (today at the left edge, **+60 d** at the right); the vertical axis is

one **orbital period** (its length in minutes is printed at the right). Each cell is shaded **yellow when the satellite is in sunlight** and left **dark when it is in Earth's shadow (eclipse)**, so the eclipse band — and the “sunline” at its edge — stands out, widening and narrowing as the orbit plane precesses relative to the Sun and vanishing entirely during **full-sun seasons** (handy for judging solar-panel charging). Below the raster a live readout shows the **current status** (**SUN/SHADOW**), the **eclipse minutes per orbit** and percentage for the current orbit, and the time to the next Sun↔shadow transition. **,// scroll one day at a time** through the 60-day window (forward indefinitely, not before today). **r** recomputes; **`** returns to Passes.

Eclipse uses a cylindrical-shadow model (no penumbra) and the raster is sampled, so the band edges and transition times are good to about a minute.

Track

The main operating screen. Top to bottom it shows: **azimuth / elevation** (and GP age), **range / range-rate** (and an orange **ECL** flag in eclipse), the selected **transponder**, the **downlink (DN) and Doppler-corrected receive (RX)** frequencies, the **uplink (UP) and transmit (TX)** frequencies, the **passband** position line, the **calibration** line, and the **radio status**. On an FM bird that needs a subaudible **PL/CTCSS tone**, a **PL nn.n Hz** line appears (orange when the radio is on and the rig supports CAT tone, grey otherwise, or **PL nn.n Hz (rig n/a)** if the selected rig can't set tone over CAT).

Controls:

- **m** — switch between **TUNE** and **CAL** modes.
- **d** — cycle the **Doppler tune mode** (linear birds): **FULL** One True Rule → **downlink-only** → **uplink-only** → **hold-both**. The passband line shows the active mode.
- **l** — **Log QSO** (also on the Polar screen): capture the contact you're working. Radio control keeps running while you fill in the entry.
- **t** — cycle to the next transponder.
- **c** — set the **CTCSS/PL tone** for this satellite (numeric entry: a tone in Hz, **0** to force it off, or blank to revert to the built-in default).
- **r** — turn radio output **on/off** (sets modes and begins Doppler service).
- **o** — turn **rotator** pointing **on/off** (if a rotator is configured; sends live az/el to the selected backend, parks on stop). See §17.
- **p** — open the **Polar** plot.
- **g / w / e** — show the **workable grids / US states / DXCC** reachable under the footprint **right now**, refreshing live; radio and rotator control keep running while you look. (The same three lists are also available per-pass from the Passes screen.)
- **f** — open **Manual mode** (frequency calculator; see below). Useful when you have no CAT-controlled radio and are tuning by hand.
- **ENTER** — save the current calibration **for this satellite**.
- **`** — first press stops the radio output; second press goes back.

TUNE-mode keys: **,//** tune down/up the passband, **s** cycle step (100/1000/5000 Hz), **x** re-center. CAL-mode keys: **,//** trim downlink, **;/.** trim uplink, **s** cycle step (10/100/1000 Hz), **x** zero. See §9 and §10.

Manual mode (f from Track)

Manual mode is for operating without a CAT-controlled radio: it shows the same live data as Track but **never commands a radio or rotator**. Instead, you fix one leg — the frequency you'll hold on your own rig — and it shows the **Doppler-corrected frequency to tune the other leg to**, updated live, with your saved calibration applied.

The two frequency rows are marked **HOLD** (the leg you park on your radio, shown at its nominal value without Doppler) and **TUNE>** (the leg you must follow, shown Doppler-corrected). Press **u** to toggle which leg is fixed:

- **Linear birds** — fixing the downlink shows the uplink to transmit (and vice versa). **,** **//** move the fixed frequency through the passband, **s** cycles the step, **x** recenters. The HOLD leg's parked value moves with the passband; the TUNE> leg follows with Doppler.
- **FM birds** — pick which leg is fixed with **u** (typically the **VHF** leg, which has little Doppler and is the one you park). The other (UHF) leg shows the Doppler-corrected frequency to chase. A hint line notes which leg is fixed and whether it's VHF.
- **Downlink-only birds** — just the computed downlink to tune your receiver to.

m toggles **CAL** (trim the same per-satellite calibration as Track), **t** cycles transponder, **l** logs a QSO, **p** opens the polar plot, **g** opens live workable grids — and **the log, polar, and grid screens all return here** rather than to Track. **ENTER** saves calibration for this satellite. **`** or **f** returns to Track.

Polar

A sky plot centered on the zenith: N/E/S/W cardinal labels, elevation rings at 30°/60°, the horizon as the outer ring. A green dot marks the satellite; a yellow glyph marks the **Sun** (when it's above the horizon). The satellite's **path across the sky for the current pass** is drawn as a cyan arc with **A** (AOS) and **L** (LOS) markers and a white arrowhead showing the **direction of travel**. When the bird is below the horizon the arc shows the **next** pass instead, and the readout gives that pass's AOS time. The right-hand readout also shows az/el, range, range-rate, **Sun az/el**, and whether the satellite is **SUNLIT** or in **ECLIPSE**. **p**, **ENTER**, or **`** return to Track.

Workable grids (g)

The 4-character Maidenhead grid squares currently inside the satellite's footprint — every grid from which a station could work the bird at the same time you can. Reached two ways:

- **g** from **Passes** — the **union** of grids covered across the selected pass (sampled about once a minute from AOS to LOS), computed once on entry.
- **g** from **Track** — the grids under the footprint **right now**, refreshed about every 3 s. Radio and rotator tracking keep running while you view it.

Grids are listed six per row in alphabetical order. The **workable count** is shown on its own cyan line at the top of the list (the header had no room for it), and when the list spills past one page that line also shows the visible window, e.g. **1370 workable (1-48)**. **;/.** scroll a row at a time and **{/}** page. Coverage is computed with a per-grid bitset, so there is **no cap** on the number of grids — it works for any amateur satellite, including high orbits (a ~2500 km bird floods roughly 4500 grids). **`** returns to whichever screen opened it.

Workable US states (w)

The US states (and DC) currently inside the satellite's footprint — the state equivalent of the workable-grids screen, reached the same ways and in the same modes:

- **w** from **Passes** — the **union** of states covered across the selected pass, computed once on entry.
- **w** from **Track** (or **Manual**) — the states under the footprint **right now**, refreshed about every 3 s, with radio and rotator tracking still running.

States are listed by their two-letter USPS code, six per row, alphabetically, with the same cyan workable-count line and `;/ . {/}` scrolling as the grids screen. Membership is decided by a point-in-polygon test against bundled **simplified** state boundaries (about 0.1°/11 km resolution), so a footprint grazing a state line may briefly claim both neighbours — fine at footprint scale, where both are in fact workable. AK, HI and DC are included. ``` returns to whichever screen opened it.

Workable DXCC (e)

The DXCC entities currently inside the footprint — the same idea again, for **DXCC chasing**, reached the same ways (**e** from **Passes** for the per-pass union, or from **Track** / **Manual** for live now). Entities are listed by common prefix (e.g. **DL**, **JA**, **VK**, **9V**), five per row, with the same workable-count line and scrolling.

Coverage and accuracy. All **340 current DXCC entities** are included, via a hybrid model: the major countries use simplified boundary polygons (so the right country is picked from the footprint geometry), while the long tail of islands and micro-entities is represented by each entity's reference coordinate from `cty.dat` and counted as workable when that point falls within the footprint (plus a small claim radius). Country borders are coarse, so a footprint near a border may list a neighbour too, and a single-point entity is claimed as a unit rather than by exact shape. Treat this as **chasing guidance** — which entities are reachable on the pass — and confirm the actual entity worked from your own log. ``` returns to whichever screen opened it.

Location

- **e** / **o** / **a** — edit latitude / longitude / altitude.
- **g** — set position from a **Maidenhead grid** square.
- **p** — enable/disable **GPS**.
- **s** — cycle the **GPS source** (Grove 9600, Grove 115200, Cap LoRa868, Cap LoRa1262).
- **c** — set the **UTC clock** manually (YYYY-MM-DD HH:MM:SS).
- ``` — back (applies the position to the predictor).

Update

- **k** or **ENTER** — update GP data from your configured source and sync the clock (NTP). The same action also refreshes the AMSAT OSCAR **activity marks** shown on the Satellites list, the **space-weather** data (solar flux + Kp), and the terrestrial **weather** for your site, so one press brings everything current. The Update screen notes this so it's clear **k** does more than GP.
- **a** — fetch and cache **all** transponders for offline use. This runs in small batches across **automatic reboots**: CardSat caches a handful of satellites, reboots to get a fresh network

connection, and continues where it left off, repeating until every satellite is done. Each reboot returns to the **Update** screen showing the running count (e.g. “Caching 24/90”), and the run finishes on “Cached all 90 transponders”. The whole pass takes a few minutes and several reboots — this is normal and lets the unit cache the full catalog reliably even on a weak Wi-Fi link. A run resumes automatically if interrupted (e.g. by a power cycle); to cancel a pending run, delete `/CardSat/tx_resume.txt` from the card. Satellites with no transmitters in the SatNOGS database are cached as an empty list, which is expected.

- `w` — connect WiFi only (no download).
- ``` — back. Diagnostics print to the serial monitor at 115200.

Settings

Settings are grouped into four submenus — **Radio / CAT**, **Rotator**, **Station / display**, and **Network / data**. `;/.` move; ENTER opens a submenu (or, inside one, edits a text field or runs an action); `,//` change an adjustable row; ``` backs out to the submenu list, then home. Press `h` anywhere for the on-screen key reference. The notable rows:

Row	Action
Radio	<code>,//</code> select model (auto-sets address + baud)
CI-V addr	ENTER → edit (hex); Icom only
CAT baud	<code>,//</code> cycle 1200...115200 (incl. 57600) — applies to all radio protocols
Min pass el	<code>,//</code> 0–30°
Decay solar	<code>,//</code> cycle assumed solar activity mean → min → max → auto for the orbital-analysis decay estimate (changes the headline number and the bracket). auto uses the live F10.7 flux fetched with GP data
Weather units	<code>,//</code> cycle the units for the Weather screen: °F, mph → °C, km/h → °C, m/s . Under <i>Station / display</i> .
WiFi SSID	ENTER → edit · s → scan for networks and pick one
WiFi pass	ENTER → edit
Save & test WiFi	ENTER → connect and report OK/FAIL
AOS alarm	<code>,//</code> or ENTER toggle on/off
Rotator (+ type / host / port / baud / ranges / offsets / deadband / park / pre-point)	<code>,//</code> adjust, ENTER edits host/port. Rot type cycles GS-232 → rotctl (net) → PstRotator → Yaesu (direct) ; see §17
GP source	ENTER → source picker : AMSAT (default), any CelesTrak JSON-PP category (Amateur Radio listed first), or Custom URL... — see §14

Row	Action
VFO Type	,// or ENTER toggle <i>Main Up/Sub Dn</i> ↔ <i>Main Dn/Sub Up</i>
Sat mode	,// or ENTER toggle the rig's satellite mode on/off
CAT rate	,// adjust the CAT update period in 10 ms steps (default 500 ms; soft-floored to what the CAT baud can service)
CAT delay	,// adjust the pause after each command, 0–200 ms in 2 ms steps (default 70 ms; CI-V/Icom only)
Dopp FM band	,// the FM-leg write deadband, 0–2000 Hz in 25 Hz steps (default 300 Hz). CardSat only re-sends an FM frequency once Doppler has moved it more than this — FM's wide passband absorbs the rest, so a loose value avoids needless CI-V chatter
Dopp linear band	,// the SSB/CW-leg write deadband, 0–1000 Hz in 10 Hz steps (default 50 Hz). Tighter than FM because linear modes need close tracking; near closest approach CardSat tightens this automatically
Dopp lead	,// the predictive-lead cap, 0–100 ms in 5 ms steps (default 50 ms; 0 = off). On fast overhead passes CardSat can compute Doppler slightly ahead to mask CAT latency, tapering the lead to zero near closest approach. Raise it if your rig's CI-V is slow; set 0 to disable
Screen sleep	,// cycle off / 30 s / 1 min / 2 min / 5 min — blanks the backlight after that idle time
My callsign	ENTER → enter your station callsign (stored uppercase); used in the log and ADIF STATION_CALLSIGN
QRZ user / QRZ pass	ENTER → enter your QRZ.com username / password for the QRZ Lookup screen (requires a QRZ XML-data subscription). Password shown masked. Under <i>Network / data</i> .
Backup config+favs → SD	ENTER → copy config + favorites to config.bak / favs.bak
Restore config+favs	ENTER → restore them from the backup files

Row	Action
Reset all data	ENTER → type ERASE to wipe everything (red row)
CAT type (+ host / port / user / pass)	, // cycle Wired CI-V → Icom LAN → rigctl (net) . Icom LAN drives the IC-9700 over RS-BA1 (see §3); rigctl drives a radio attached to a remote Hamlib rigctld server over TCP (host/port, default 4532). Under <i>Radio / CAT</i> .
Rigctld server (+ port)	, // enable a rigctld server so a PC (Gpredict, WSJT-X, a logger) drives the radio through CardSat over TCP (default 4532); VFOA=downlink, VFOB=uplink. Under <i>Radio / CAT</i> .
Rotctld server (+ port)	, // enable a rotctld server so a PC drives the wired GS-232 rotator through CardSat over TCP (default 4533). Under <i>Rotator</i> .
Rotator: manual control	ENTER opens a screen to jog az/el by hand with live read-back. For a Yaesu (direct) rotator this is also where you calibrate the ADC : it shows live ADC counts and you capture the axis endpoints with 1/2/3/4 (az 0 / az full / el 0 / el full) — see §17 and ROTOR_INTERFACE.md . Under <i>Rotator</i> .

WiFi scan

Reached from **Settings** by pressing **s** on the **WiFi SSID** row. CardSat scans for nearby networks and lists them strongest-signal first, with each network's RSSI (dBm) and a * for secured networks.

- ;/. — select a network.
- **ENTER** — use it: the SSID is saved and you're taken to password entry (open networks skip the password and return to Settings).
- **r** — rescan.
- **`** — back to Settings.

Edit

A simple text entry box (for the fields above and manual GP/transponder/time entry). Type to append, **DEL** to backspace, **ENTER** to confirm, **`** to cancel.

Adding a manual satellite (from Satellites → **n**): you'll be prompted in order for the **name**, **NORAD ID**, **epoch** (YYYY-MM-DD HH:MM:SS, UTC), then the GP mean elements — **inclination**, **RAAN**, **eccentricity** (e.g. 0.0006190), **argument of perigee**, **mean anomaly**, **mean motion** (rev/day), and **BSTAR** (drag; 0 is fine if unknown). The satellite is stored with the downloaded ones and persists across GP refreshes.

Adding a manual transponder (from Passes → n): you'll be asked, in order, for **Downlink low (Hz)**, **Uplink low (Hz, 0 = none/beacon)**, **Downlink high (Hz, 0 = single channel/FM)**. If you gave a downlink high above the low *and* an uplink, it's treated as **linear** and you'll also be asked **Uplink high (Hz, 0 = same bandwidth)**, **Inverting? (y/n)**, and finally the **Mode**. Single- channel entries skip straight to Mode. Manual transponders are stored separately from the SatNOGS cache so a GP/transponder refresh won't erase them. Up to **64 transponders** are held per satellite (SatNOGS plus your manual ones) — enough for even the busiest birds, such as the ISS (~50 transmitters on SatNOGS).

09

Doppler tuning and the One True Rule

A satellite's motion shifts both the frequency you **receive** (downlink) and the frequency the satellite **receives from you** (uplink). CardSat corrects both, continuously.

CardSat follows the AMSAT “**One True Rule**” (Paul Williamson, KB5MU): *tune both the transmitter and the receiver to achieve a constant frequency at the satellite*. In practice the firmware holds your chosen spot in the transponder fixed **as the satellite sees it**, and applies the Doppler correction to **both** the receive and transmit frequencies every cycle. The result: you tune somebody in, let go, and nobody drifts through the passband.

The corrections are:

$RX = \text{downlink} \times (1 - \beta) + \text{calDL}$	(tune your receiver here)
$TX = \text{uplink} \div (1 - \beta) + \text{calUL}$	(transmit here)
$\beta = \text{range-rate} \div c$	(c = speed of light)

where **downlink/uplink** are the satellite-side frequencies of your chosen passband spot, and **calDL/calUL** are your calibration offsets (§10).

Choosing your spot in the passband

On a **linear transponder** you can move through the passband two ways:

- **TUNE mode** (device keys): press **m** until the passband line shows **<TUNE>**. **,**/**/** move your operating point down/up; **s** cycles the step; **x** recenters. The **PB** line shows your offset from band center, the half-width, and **INV** for inverting birds.
- **Radio-knob mode** (One True Rule, the natural way): press **d** to cycle the tune mode. In **FULL** (passband line shows **<FULL>** in orange) just **turn the radio's tuning knob** to move around the passband — CardSat reads your downlink, works out where you are, and keeps both legs Doppler-corrected around that fixed satellite frequency. Let go and you stay put. Pressing **d** cycles on through **downlink-only** (**<DL>** — One True Rule on the downlink, uplink left alone), **uplink-only** (**** — only the transmit leg is corrected; handy when an SDR or second receiver handles the downlink, and it needs no frequency read-back so it works even on set-only rigs), and back to **hold-both** (**<TUNE>**, device-key tuning). **FULL** and downlink-only need a rig that reports frequency; the cycle skips them otherwise.

For an **inverting** transponder the uplink moves opposite to the downlink (tune the downlink up, the uplink goes down); for a non-inverting one they track together. CardSat handles the mapping automatically.

Sidebands

For linear transponders CardSat sets the rig's modes for you: **USB on the downlink, LSB on the uplink** (because an inverting linear transponder flips the spectrum, so an LSB uplink is heard as a normal USB downlink). The exception is any bird with an uplink or downlink

below 30 MHz (HF), which uses **USB up and USB down**. FM and single-channel birds use the transponder's own mode on both legs.

10

Calibration

Real radios and satellites have small oscillator offsets. **CAL mode** lets you null them out per satellite.

1. On **Track**, press **m** until the cal line shows **<CAL>**.
2. Find a known signal (your own downlink, or the satellite's beacon).
3. Trim the **downlink** with **,//** and the **uplink** with **;/.. s** cycles the step (10/100/1000 Hz); **x** zeroes both.
4. Press **ENTER** to save the calibration **for this satellite** — it's stored and reloaded automatically next time you track that bird.

The passband position is *not* persisted (it's per-pass); calibration *is*.

Editing calibrations on the SD card

Per-satellite calibrations are stored as a plain-text file, so you can author or bulk-edit them on a computer instead of nudging each bird by hand. With the microSD removed (or over USB mass storage), open **/CardSat/calib.txt**. Each line is one satellite:

```
# norad  downlink_Hz  uplink_Hz      (lines starting with # or ; are comments)
43017   -250    300
25544    120     0
```

- The first field is the **NORAD catalog number**; the next two are the **downlink** and **uplink** offsets in **Hz** (signed — negative lowers the frequency). These are the same **calDL/calUL** values you trim in **CAL** mode.
- Whitespace-separated, one satellite per line. Blank lines and lines beginning with **#** or **;** are ignored, so you can annotate the file freely.
- The file is read each time you select or track a satellite, so edits take effect the next time you open that bird — **no reflash needed**. Saving a calibration on the device (**CAL** → **ENTER**) rewrites this file but preserves your comment lines.

CTCSS tone overrides work the same way in **/CardSat/tones.txt**, one line per satellite as **norad tone_tenths** (tenths of a Hz, so **670** = 67.0 Hz; **0** forces the tone off):

```
# norad  tone_tenths
25544    670
```

Both files live on the microSD card if one is present, otherwise in the device's internal flash. If a satellite has no line in **calib.txt**, the global calibration from **Settings** is used.

11

Working a pass, step by step

An FM bird (e.g. a repeater satellite):

1. **Update** → **k** to refresh GP data (and the clock) if you have WiFi.
2. **Satellites** → highlight the bird → **f** to favorite it → **ENTER** to open Passes.
3. Pick the pass you want; press **d** to preview its elevation curve if you like.
4. At AOS, **ENTER/t** to Track. Press **r** to start sending to the radio.
5. CardSat sets FM on both legs and Dopplers the single channel. Talk.

A linear (SSB/CW) bird:

1–4 as above. On Track you'll start in **TUNE** (press **d** to cycle to FULL radio-knob mode).
 5. Press **r** to enable radio output. CardSat sets **USB down / LSB up** (or USB/USB on HF) and corrects both legs. 6. Tune to a clear spot — with the device **,//** keys, or by turning the rig's knob in **<FULL>** mode. Your downlink stays put; the uplink tracks (inverted if the bird inverts). 7. If your own signal isn't centered, switch to **CAL (m)**, trim, and **ENTER** to save.

Watch the **ECL** flag and the polar Sun glyph — some birds change behavior in eclipse.

12

AOS alarm and deep sleep

AOS alarm (toggle in Settings → *AOS alarm*): when enabled and you have favorites and a set clock, CardSat tracks the soonest upcoming favorite AOS in the background. It **beeps** at T-60 s, T-30 s, and T-10 s, then sounds a longer double-beep and shows a blinking “**AOS!**” banner at acquisition. Once a pass is underway it also chirps at **TCA** (closest approach / peak elevation — a double mid-tone) and at **LOS** (a descending two-tone), so you can follow a pass by ear without watching the screen. All of these sounds are governed by this one setting: turning the AOS alarm off silences the AOS, TCA and LOS cues together. A small orange countdown banner appears on any screen within the last minute.

Deep sleep (Next Passes → *z*): CardSat computes the next favorite AOS and puts the ESP32 into deep sleep until **~60 s before** it, dramatically extending battery life between passes. The screen shows how long it will sleep and for which satellite. On wake, the unit reboots straight to **Next Passes** with the pass imminent and the alarm ready. (Press the reset button to wake early.) The clock is preserved across deep sleep, but make sure it was set — via NTP or GPS — before sleeping.

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Sun and eclipse

CardSat computes the Sun's position and whether the satellite is illuminated:

- **Polar screen** — a yellow Sun glyph at its az/el (when above the horizon), the Sun's azimuth/elevation in the readout, and **sat SUNLIT** (green) or **sat ECLIPSE** (orange).
- **Track screen** — an orange **ECL** flag when the satellite is in Earth's shadow.
- **Pass detail** — the elevation curve is colored yellow (sunlit) / blue (eclipse), and a sunlit-percentage is given for the pass.

This uses a low-precision Sun ephemeris and a cylindrical Earth-shadow model — the sunlit↔eclipse transition is a hard edge rather than the few-second penumbral fade, which is plenty for knowing whether a bird has power or is optically visible.

Sun / Moon antenna tracking

Sun / Moon on the main menu shows the live position of both bodies from your location. It opens in a **graphical sky view**: a polar dome (zenith at centre, North up, horizon at the rim) with the Sun drawn as a rayed yellow disc and the Moon as a cyan crescent, so you can see at a glance where each one is. A compact panel on the right lists azimuth, elevation and above/below-horizon for both. A body below the horizon is shown faintly just outside the rim so its bearing is still readable. Press **g** to toggle between the graphic and a plain azimuth/elevation data list. **;/** selects which body the rotator follows (marked with a green ring, not a highlight bar). Press **o** to drive the rotator at the selected body — useful for antenna gain checks against Sun noise, EME pointing, or rotator calibration against a visible target. **x** stops; **`** parks and disengages.

Behavior notes:

- The rotator has **one master at a time**: engaging Sun/Moon tracking disengages satellite rotator tracking and vice versa, and opening **Rotator manual** control disengages both.
- While the selected body is **below the horizon** the rotator parks and waits; tracking resumes automatically when it rises.
- Tracking keeps running if you navigate to other screens — an orange **SUN / MOON** tag in the header shows it's active. Going back from the Sun/Moon screen parks and disengages.
- The Moon ephemeris is a low-precision series (~arc-minutes after topocentric parallax correction) — far finer than any amateur antenna beamwidth. The Sun is good to ~0.01°.

Space weather

Space Wx on the main menu summarises the indices that matter most for propagation: the **solar 10.7 cm radio flux** (F10.7, a proxy for solar activity and ionospheric ionisation), the **planetary Kp index** (geomagnetic disturbance, 0–9), and the **running A index** (the daily-equivalent geomagnetic amplitude, shown when the feed provides it). They're fetched from NOAA SWPC together with GP updates, or on demand

with **r** (needs WiFi). The Kp and A fetch is independent of the flux fetch, so a hiccup in one never suppresses the other. Each value is labelled in plain terms — flux low/moderate/good/very-high, Kp quiet/unsettled/minor-storm/moderate-storm/major-storm, and A quiet/unsettled/active/storm — and colour-coded, with a short **operating outlook** line translating the numbers into what to expect on HF and satellite paths, plus a note of how old the data is.

This is a planning cue, not a forecast: the flux and Kp are observed values, and the outlook text is a simple heuristic reading of them, not a calibrated propagation prediction. A high Kp (storm) is the main thing to watch — it warns of auroral flutter on VHF and disturbed high-latitude HF.

Weather

Weather on the main menu (just below Space Wx) shows current conditions and a short forecast for your operating site — handy for portable and field operation. It displays the current temperature and sky condition, wind speed/direction and humidity, then a row for each of the next few days with the day's condition, high/ low, and chance of precipitation.

The data comes from **Open-Meteo** (open-meteo.com), a free, no-key weather service. The location is taken from the same site coordinates the prediction engine uses (your GPS fix or manually set lat/lon), so set your location first. Weather is fetched when you run **Update**, and also refreshes automatically on entry to the screen if WiFi is already connected; **r** forces a refresh. Like Space Wx, the last result is cached to flash, so it remains viewable offline with a note of its age.

Units (°F·mph, °C·km/h, or °C·m/s) are selectable in *Settings* → *Station / display* → *Weather units*; changing them re-labels the cached values immediately without needing a re-fetch.

Weather data by Open-Meteo.com, licensed under CC BY 4.0.

Transponder database

From the **Satellites** list, press **t** to browse every transponder and beacon entry the on-device catalog holds for the selected satellite (sourced from SatNOGS with the transponder cache). Each entry is shown as a short block: its description, the **downlink** (a range for linear transponders) with mode, and the **uplink** with any CTCSS tone and inverting/linear flags. **;/** scrolls through the list. It's a quick offline reference for a bird's frequencies and modes — handy for checking what a satellite carries without a radio connected. If nothing shows, the transponders haven't been cached yet; run **Update** with WiFi on.

QRZ callsign lookup

QRZ Lookup on the main menu looks up a callsign in the **QRZ.com** database and shows the operator's name, mailing address, country, grid square and licence class. It uses QRZ's XML data service, which **requires a QRZ XML-data subscription**.

To use it, enter your **QRZ username** and **password** in *Settings* → *Network / data* (rows **QRZ user** / **QRZ pass**). Then open **QRZ Lookup**, press **ENTER**, type a callsign, and press **ENTER** again. CardSat logs in to QRZ, caches the session key, and displays the result; the key is reused for subsequent lookups until it expires (then it re-logs in automatically).

The screen handles the obvious cases plainly:

- **No WiFi** — it simply says WiFi isn't connected; connect via Update or Settings and try again.
- **No credentials** — it explains that a QRZ XML subscription is required and that you need to enter your username and password in Settings.
- **Not found / login error** — the QRZ error message (e.g. "Not found", "password incorrect") is shown in the status line.

Your QRZ password is stored on the device the same way the WiFi and radio-LAN passwords are; it is shown masked in Settings. CardSat talks to QRZ over HTTPS.

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GP age and accuracy

Theory — what GPs and SGP4 actually are

Why a satellite needs “elements” at all. You cannot just store a satellite’s position and velocity and expect it to stay useful — it sweeps through space at ~ 7.5 km/s and the orbit itself slowly changes shape. Instead, the satellite is described by a compact set of numbers that, fed to a matching math model, *regenerate* the position at any time. Those numbers are the **orbital elements**; the model is **SGP4**. The two are a matched pair: the elements only mean what they mean *because* SGP4 is the model that will consume them.

TLE, OMM and “GP”. The historical packaging is the **Two-Line Element set (TLE)** — two 69-character lines, a fixed-column format dating to punch cards, holding the elements plus epoch, drag term, and catalog bookkeeping. **OMM** (Orbit Mean-elements Message) is the modern, self-describing replacement carrying the *same* numbers as JSON, XML or KVN instead of fixed columns. CelesTrak’s umbrella term for “current mean elements in whatever format” is **GP** (General Perturbations). CardSat downloads **GP data as JSON/OMM** (it never has to parse the brittle column format off the wire), but internally it renders each set back into a TLE line-pair — epoch encoded as **YYDDD.d**, B* in TLE exponent notation, with the line checksums — and hands that to the SGP4 library, which still ingests elements the classic way. The element values are identical; only the wrapper changes.

What SGP4 is. SGP4 (Simplified General Perturbations 4) is the analytic propagator that NORAD/USSPACECOM elements are *fit to*. “Analytic” means it isn’t numerically integrating forces step by step; it applies closed-form expressions for the perturbations that matter for a near-Earth satellite over days-to-weeks:

- **Earth’s oblateness** — the equatorial bulge (the **J2** zonal harmonic, with smaller J3/J4 terms) that precesses the orbit plane and rotates perigee. This is the dominant non-Keplerian effect and the reason pass times march earlier each day.
- **Atmospheric drag** — modelled through the **B*** (“B-star”) term, a fitted ballistic-and-density coefficient that slowly shrinks the orbit. (A companion model, **SDP4**, extends the same scheme to deep-space orbits with period > 225 min by adding lunar/solar gravity and resonance terms — the pair is often written “SGP4/SDP4.” In practice CardSat’s birds are all near-Earth, so the SGP4 branch is what runs.)

Crucially, the **“mean” elements are not osculating elements**. They are deliberately *detuned* values chosen so that, after SGP4 adds its periodic terms back in, the output matches reality. You cannot mix them with a different model, average them, or read the mean motion as an instantaneous rev-rate and expect physical truth — they are model-specific fitting constants. This is also why CardSat propagates with the **WGS72** gravity constant set: that is the set the elements were fit against, and using WGS84 would introduce a small systematic error.

From element to az/el — the frames. Propagating gives position and velocity in the **TEME** frame (True Equator, Mean Equinox — the quirky inertial-ish frame SGP4 outputs). To point an antenna, CardSat rotates the observer’s geodetic latitude/longitude/altitude into the same TEME frame using **GMST** (Greenwich Mean Sidereal Time — Earth’s rotation angle for the instant), differences the two position vectors to get the slant vector, and resolves that into

topocentric **azimuth/elevation**. Range-rate (for Doppler) is taken directly from the projection of the relative *velocity* onto the slant direction, evaluated at the exact fractional second rather than by differencing whole-second range samples — cleaner right at closest approach where range-rate changes fastest.

Where the error comes from. SGP4 near a fresh epoch is good to roughly a kilometre. Error grows with **element age** because the small unmodelled accelerations (drag fluctuations from space weather, higher-order gravity, solar radiation pressure) integrate over time, and drag is the worst offender — it depends on upper-atmosphere density, which swings with solar activity and isn't known in advance. So a low, draggy orbit (ISS, cubesats) can develop noticeable along-track timing error (the satellite arriving early or late along an otherwise-correct path) within a week or two, while a high, stable orbit stays usable far longer. Along-track error shows up to you as **pass-time slip**; cross-track error shows up as **pointing/Doppler error**. The practical defence is simply to refresh elements often enough for the orbit you care about.

GP age indicator

SGP4 predictions degrade as the GP mean elements age. CardSat shows the **age of the element set** (days since the GP epoch) on the Passes and Track screens, color coded:

- **Green** — under 14 days (fresh).
- **Yellow** — 14–28 days (getting old).
- **Red** — over 28 days (stale; expect timing/pointing error). In the Next Passes schedule, stale sats are flagged with a red !.

Refresh with **Update → k** whenever you have WiFi. For low orbits, weekly (or fresher) elements are best.

Choosing the element source

By default CardSat pulls the **AMSAT** daily GP bulletin (all amateur satellites, JSON, including the friendly **AMSAT_NAME**). To change source, open **Settings → GP source** and press ENTER for a picker:

- **AMSAT (amateur)** — the default bulletin.
- **CelesTrak categories** — any CelesTrak GP group in JSON-PP format. Amateur Radio is listed first, followed by SatNOGS and the Special-Interest, Weather & Earth Resources, Communications, Navigation, Scientific and Miscellaneous groups. CelesTrak omits **AMSAT_NAME**, so CardSat falls back to **OBJECT_NAME** and the data parses correctly.
- **Custom URL...** — type any URL that returns OMM JSON / JSON-PP.

Move with **;/.** (or **{/}** to page) and press ENTER to select. The choice is saved immediately and used by the next **Update → k**.

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Working offline

CardSat is designed to operate with no network in the field:

1. With WiFi, go to **Update** and press **k** (GP + clock) and then **a** (cache **all** transponders). Both are written to flash. The transponder cache runs in batches across several automatic reboots and finishes on “Cached all N transponders” — let it run to completion before going offline.
2. After that, everything — pass prediction, transponders, Doppler, the schedule — works with WiFi off. The clock keeps running (and survives deep sleep); use GPS or manual **c** entry to set it where there’s no NTP.

Cached data persists across power cycles until you refresh it or perform a reset.

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Radio-specific notes

CardSat speaks three CAT dialects, selected by the **radio model** in Settings. Defaults are editable; the **CI-V address** field applies to **Icom only**.

Radio	Family / protocol	Default baud	Interface	Read-back
IC-820	Icom CI-V	9600	CI-V 5 V single-wire	yes
IC-821	Icom CI-V	9600	CI-V 5 V single-wire	yes
IC-910	Icom CI-V	19200	CI-V 5 V single-wire	yes
IC-970	Icom CI-V	9600	CI-V 5 V single-wire	yes
IC-9100	Icom CI-V	19200	CI-V 5 V single-wire	yes
IC-9700	Icom CI-V	19200	CI-V 5 V single-wire	yes
FT-847	Yaesu 5-byte	57600	serial (TTL/RS-232)	yes ¹
FT-736R	Yaesu 5-byte	4800	serial (TTL/RS-232)	no
TS-790	Kenwood ASCII	4800	RS-232 (MAX3232)	yes
TS-2000	Kenwood ASCII	57600	RS-232 (MAX3232)	yes

Protocol command sets follow the Hamlib backends ([icom](#), [yaesu/ft847](#), [yaesu/ft736](#), [kenwood/ts2000](#), [kenwood/ts790](#)). Serial framing is set automatically: **Icom 8N1**, **Yaesu 8N2**, **Kenwood 8N1** — **except 8N2 at 4800 baud**. The TS-790 generation (IF-232C interface: TS-450/690/790/850/950) requires **two stop bits at 4800 baud** (one stop bit at higher rates such as the TS-2000's 57600); CardSat selects this by baud automatically. Note the TS-790's CAT is via the **optional IF-232C** adapter — its operating manual documents the interface but not the command set, so the TS-790 mapping leans on the shared Kenwood protocol and is the least bench-verified of the rigs here.

The CAT *interface circuit* differs by family. **Icom CI-V** uses a 3.3 V-safe single-wire interface ([CIV_INTERFACE.md](#)); **Yaesu** and **Kenwood** use RS-232-level serial — build a **MAX3232** level shifter as in [RS232_INTERFACE.md](#) (! untested; build at your own risk).

Automatic PL / CTCSS tone for FM satellites

Several FM birds require a subaudible **PL (CTCSS) tone** on the uplink. CardSat applies it automatically: when the active transponder is **FM with an uplink** and the satellite is in its built-in tone table, it enables the rig's **TX CTCSS encoder** at the right frequency the moment radio output is on, turns it **off** again when you switch to a transponder that doesn't need one, and disables it when you turn radio output off (so your rig isn't left transmitting a tone). The tone in use is shown on the Track screen as a **PL nn.n Hz** line, and a status flash confirms it (PL 67.0 Hz on uplink).

Built-in tones (operating tone, by NORAD id): **ISS** 67.0, **SO-50** 67.0, **AO-91** 67.0, **AO-92** 67.0, **PO-101** 141.3. SatNOGS carries no structured tone field, so this list lives in [SatDb::knownCtcssHz\(\)](#) — extend it there as new FM satellites appear. **SO-50 note:** the 67.0 Hz figure is the *operating* tone; arming its 10-minute timer with a 74.4 Hz burst is a separate manual step on your radio.

Setting tones yourself. Press **c** on the Track screen to set a tone for the current satellite. Enter a frequency in Hz (e.g. **67.0**, **141.3**) and it is snapped to the nearest standard CTCSS tone; enter **0** to force the tone **off** for that bird; leave it **blank** to clear your override and fall back to the built-in default. The override is stored per satellite (by NORAD, in **tones.txt**) and wins over the table, so it survives reboots and a GP/transponder refresh, and it's how you add a tone for any FM bird not already in the built-in list.

Tone-over-CAT is supported on **IC-910/9100/9700**, **FT-847**, and **TS-2000** (the **hasTone** flag in **radio_profiles.h**); on other models the **PL** line reads (**rig n/a**) and no tone command is sent. Encoders by family: **Icom** CI-V **1B 00** (tone freq, BCD) + **16 42** (encoder on/off); **FT-847** sat-TX opcodes **2B** (tone, via the CAT code table) + **4A...2A/8A...2A** (encoder on/off); **TS-2000** **TN** (tone number) + **T0** (encoder on/off). All are taken from the Hamlib backends; like the rest of CAT they're host-verified but not yet exercised on a real radio — watch the serial trace and confirm the rig shows the tone.

Icom. CardSat drives MAIN/SUB directly and forces the rig's built-in satellite mode **off** at startup on the rigs that expose it over CAT (IC-910, IC-9100, IC-9700). Downlink on SUB, uplink on MAIN. Read-back uses **0x03** and works on all six (including the IC-820/821/970, per Hamlib). Wrong-VFO fixes live in **radio_profiles.h**.

IC-910 satellite-mode & tone commands differ from the IC-9100/9700. The IC-9100/9700 toggle satellite mode with CI-V **0x16 0x5A**, but the **IC-910 uses a different command group entirely: 0x1A 0x07** (verified from the IC-910 CONTROL COMMAND table — on the IC-910, command **0x16** has no satellite-mode sub-command). Likewise the CTCSS tone encoder is **0x16 0x42** (Repeater tone) on the IC-9100/9700 but **0x16 0x43 (Subaudible tone) on the IC-910** — its **0x42** is the auto-notch filter. CardSat sends the right command for each rig. (Earlier firmware sent **0x16 0x07/0x16 0x42** to the IC-910, so sat mode never engaged and the tone key toggled the notch; both are fixed.)

IC-820H vs IC-821H MAIN/SUB select. These two rigs use the *same* CI-V command (**07**) for band select but with the **two sub-commands swapped** — a quirk confirmed from each radio's own manual (CI-V command table):

Radio	Address	Main band access	Sub band access
IC-821H	4C	0x07 D0	0x07 D1
IC-820H	42	0x07 D1	0x07 D0

CardSat ships the correct (reversed) mapping for each, so both tune the right VFO out of the box. If you ever swap one rig's CI-V address onto the other's profile, remember the band-select bytes differ too. The frame is otherwise identical: preamble **FE FE**, controller **E0**, command **07**, end **FD**.

Icom over the network. The **IC-9700** can be driven over WiFi/Ethernet instead of the CI-V bus — set **CAT type** → **Icom LAN** in Settings (with the IC-9700 selected as the radio model). CardSat carries the *same* CI-V frames shown below inside the RS-BA1 UDP protocol, so MAIN/SUB, read-back, sat mode and CTCSS behave identically; only the transport changes. Other networked Icoms (IC-705, IC-7610, IC-785x) use the same RS-BA1 protocol so the link would establish, but they are single-receiver radios without the MAIN/SUB satellite architecture and are not supported. Setup is in §3; the wire protocol is documented in **ICOM_LAN_PROTOCOL.md**.

Yaesu (FT-847, FT-736R). 5-byte CAT (four data bytes + opcode), big-endian BCD at 10 Hz. CardSat enables CAT at startup and sets the satellite RX (downlink, opcode `0x11`) and TX (uplink, `0x21`) VFOs directly.

¹ **FT-847 read-back** uses “read freq & mode” (`0x03`, patched to `0x13` for the SAT-RX/downlink VFO), which returns 4 big-endian BCD bytes + a mode byte. This works only on **firmware-updated** FT-847s — early units have no read capability and stay silent (CardSat times out and holds the passband). In satellite mode the radio can occasionally return the uplink VFO instead (Hamlib #1286); CardSat rejects any read that jumps more than 1 MHz from the commanded downlink, so a stray reply holds steady rather than jerking the passband. So **radio-knob (One True Rule) tuning works on a firmware-updated FT-847**; on older units, use the device TUNE keys.

The **FT-736R** can’t report frequency over CAT at all (TUNE keys only), and its native opcodes differ from the FT-847 — the proven path is an **FT-847-emulating** interface (KA6BFB / HS-736USB), in which case select **FT-847**. Put either radio in **VFO** (and its satellite mode in VFO) before enabling CAT.

Kenwood (TS-790, TS-2000). ASCII commands terminated by `;`: downlink on **VFO A** (`FA<11-digit Hz>;`), uplink on **VFO B** (`FB...;`), mode via `MD<n>;`, read via `FA;`. Connects over **RS-232** (DB-9) — use a **MAX3232** level shifter, not the CI-V circuit. On the TS-2000, bridge **CTS/RTS** (or use “RTS +12 V”) for its handshake quirk and verify the VFO-A/B ↔ main/sub-band behaviour for your firmware. The TS-790 supports a subset of the same commands.

All Yaesu/Kenwood sat rigs: CAT cannot switch the band pair. You select the uplink/downlink bands and engage the rig’s own satellite / full-duplex mode **on the radio**; CardSat Doppler-tunes within that. (Same as SatPC32.)

Watching the CAT trace

CardSat prints **every CAT frame it sends** to the serial monitor at **115200 baud**, decoded, with the radio’s reply where the protocol provides one. Connect over USB, open a serial monitor at 115200, and you’ll see traffic like:

Icom (CI-V):

```
[CI-V TX] FE FE A2 E0 07 D1 FD sel-band SUB
[CI-V TX] FE FE A2 E0 05 00 60 58 45 14 FD set-freq 145456000 Hz
[CI-V RX] radio ACK (FB)
[CI-V TX] FE FE A2 E0 03 FD read-freq req
[CI-V] SUB freq read: 145456000 Hz
```

Yaesu / Kenwood:

```
[CAT TX] 14 59 00 00 11 set-freq RX 1459000000 Hz
[CAT TX] 43 59 00 00 21 set-freq TX 4359000000 Hz
[CAT TX] FA001459000000;
[CAT TX] MD2;
[CAT RX] FA001459000000;
[CAT] VFO-A (downlink) read: 1459000000 Hz
```

How to read it:

- **[CI-V TX] / [CAT TX]** — a frame sent to the radio (raw bytes, or the ASCII string for Kenwood) plus a decode: which VFO, set-freq **<Hz>**, set-mode, read request, or CAT-on.
- **[CI-V RX] radio ACK (FB)** — Icom accepted the command; **NAK (FA)** means it rejected it. (Yaesu set commands aren't acknowledged.)
- ... **freq read: <Hz>** — the frequency read back (Icom/Kenwood) used by radio-knob One True Rule tuning; **no valid reply** if it failed.

This is the fastest way to confirm your **wiring, model, address, and baud**, and to watch the **Doppler corrections** stream during a pass. The **Icom LAN** backend adds its own **[NET]** traces for the connect/auth handshake and keepalives (the CI-V frames it carries are the same ones shown above). Silence a backend by setting **CIV_DEBUG / YAESU_DEBUG / KW_DEBUG / ICOMNET_DEBUG 0** at the top of **civ.cpp / yaesu.cpp / kenwood.cpp / icomnet.cpp** and re-build.

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Antenna rotator (GS-232, rotctl, PstRotator, Yaesu direct, rotctld server)

v0.9.8 note. The network *client* backend is now labelled **rotctl** in Settings — it connects to a rotctld server, it is not itself the daemon. Separately, CardSat can run its **own rotctld server** (Settings → Rotator → *Rotctld server*, default port 4533) so a networked PC (Gpredict, ...) drives the **GS-232 rotator wired to CardSat**; a **P** from the PC disengages CardSat's own tracking. A **manual control** screen (Settings → Rotator → *Rotator: manual control*) jogs az/el by hand with live read-back. At **el range 180°** the *flip* decision is now made once per pass and held to LOS (it previously never triggered). **rotctld/rigctld** servers have no authentication — trusted LAN only.

CardSat can drive an az/el antenna rotator through one of three interchangeable backends, chosen in Settings with **Rot type**:

- **GS-232** – a directly-attached controller speaking the **Yaesu GS-232A/B** protocol (a Yaesu G-5500 with a GS-232B, a SPID controller, or any GS-232 emulator: K3NG / RadioArtisan / ERC). Because all three ESP32-S3 UARTs are already in use (USB, CAT, GPS), this link is created by an **I2C->UART bridge** (SC16IS750/752) on a second I2C bus, so it coexists with the radio and GPS.
- **rotctld (net)** – a **Hamlib rotctld server** anywhere on the same WiFi network. CardSat is the TCP client (the same role Gpredict plays), so any rotator Hamlib supports can be driven over the LAN with no extra CardSat wiring.
- **PstRotator (net)** – a **PstRotator** (YO3DMU) instance on the LAN, driven over **UDP** (the same datagrams Gpredict/SatPC32 send it). CardSat sends **<PST>...</PST>** control messages to PstRotator's UDP Control port; PstRotator then drives whatever controller it is configured for.
- **Yaesu (direct)** – a Yaesu **G-5500-class** az/el controller wired **straight to CardSat**, with no GS-232 box: an I2C **ADS1115** reads the position-feedback voltages and an I2C **PCF8574** drives the four direction lines, both on the same I2C bus the GS-232 bridge would use. CardSat closes the pointing loop itself. This is a hardware build with its own calibration step – see [ROTOR_INTERFACE.md](#) (! untested; build at your own risk). Calibrate from **Settings → Rotator → Rotator: manual control** (capture keys 1/2/3/4 at the axis endpoints).

Only one rotator is active at a time. The pointing logic – alignment offsets, deadband, flip mode, park-on-LOS – is identical for all of them; **Rot type** only changes how the final azimuth/elevation reaches the hardware.

Wiring

Chain: **Cardputer I2C (Wire1) -> SC16IS750 -> MAX3232 -> DB-9 -> GS-232 controller.**

- The bridge runs on the Cardputer-ADV expansion I2C bus, **G8 = SDA / G9 = SCL** (**ROT_I2C_SDA / ROT_I2C_SCL** in `config.h`), on a second I2C controller (Wire1) so it

never touches the keyboard/IMU bus. These are confirmed from the M5Stack Cap LoRa-1262 pinmap and don't collide with CAT (G1/G2), the GPS UART (G13/G15), the LoRa SPI (G3/G4/G5/G6/G14/G39/G40), or the SD card (CS G12).

- GS-232 uses RS-232 voltage levels, so a **MAX3232** sits between the bridge's TTL TXD/RXD and the controller's DB-9 (three wires: TXD, RXD, GND).
- The SC16IS750 needs a 14.7456 MHz crystal (the common breakouts have one); set **ROT_XTAL_HZ** if yours differs, and its I2C address (A0/A1 strap) in **ROT_I2C_ADDR** (default 0x4D).
- On the Cap LoRa-1262, G8/G9 is exactly the I2C bus broken out to the cap's **HY2.0-4P Grove Port.A**, so a Grove SC16IS750 bridge plugs straight in. That bus also carries the cap's PI4IOE5V6408 IO expander (~0x43/0x44, used only for the LoRa RF switch, which CardSat never drives), so keep **ROT_I2C_ADDR** (default 0x4D) clear of those — or add a TCA9548A mux if anything clashes.

Settings

Enable and tune the rotator in **Settings** (scroll past the radio rows):

Setting	Meaning
Rotator	off / on (builds the selected backend)
Rot type	GS-232 (I2C bridge), rotctld (net) (TCP), or PstRotator (net) (UDP)
Net host	server IP / hostname (rotctl or PstRotator backend)
Net port	server port – rotctld TCP 4533 , PstRotator UDP 12000
Rot baud	GS-232 serial speed (1200 / 4800 / 9600); GS-232 backend only
Rot deadband	degrees; suppress moves smaller than this (anti-chatter)
Rot park az	azimuth the rotator parks at on LOS / when disabled
Rot pre-point	lead before AOS to pre-aim at the rise bearing (off / 30 s / 1-5 min)
Rot Az offset	added to commanded azimuth (mount alignment)
Rot El offset	added to commanded elevation
Rot az range	azimuth travel: 0..360 (default), -180..+180 (centred on N), or 0..450 (90 deg overlap)
Rot az lookahead	, // 0–10 s (default 3 s; 0 = off). On a 0..450 rotator only: CardSat predicts the bearing this many seconds ahead and, when a pass is about to cross north, commits early to the 361–450° overlap band so it makes a short move instead of unwinding ~360°. Has no effect on 0..360 or flipped passes. Tune to your rotator's slew speed

Rot el range	90 deg , or 180 deg = flip over the top for high passes
--------------	---

Rot type and Net port adjust in place with ,//; Net host and Net port also open a text editor with ENTER.

Using it

On the **Track** screen press **o** to start/stop pointing. The status line shows **Rot ON / off / n/c** (*n/c* = no link: the I2C bridge wasn't found, or the rotctld server isn't reachable). Pressing **o** re-attempts the link on the spot, so you can engage as soon as the controller or server comes up. While on, CardSat sends the satellite's azimuth and elevation about once a second – the GS-232 **W aaa eee** command, rotctld's **P <az> <el>**, or PstRotator's **<PST><AZIMUTH>..</AZIMUTH><ELEVATION>..</ELEVATION></PST>** datagram – but only when the position has moved past the deadband, since rotators are slow and mechanical. When the satellite sets, or you press **o** again or leave the screen, the rotator **parks** at the configured azimuth/elevation.

Pre-positioning before AOS. A slow rotator can take most of a minute to slew across the sky, so by the time it reaches the rise bearing the pass has already begun. With **Rot pre-point** set to a lead time (default **2 min**; **off** disables it), CardSat predicts the next pass for the tracked satellite and, once AOS is within that window, aims the rotator at the **rise azimuth on the horizon** instead of the idle park position – so it is already pointed when the satellite appears and live tracking takes over smoothly. Set the lead a little longer than your rotator's worst-case slew time. Outside the window it still rests at **Rot park az**.

Azimuth range (0-360 / +/-180 / 0-450). By default CardSat assumes the azimuth axis runs **0 to 360 deg**, 0 deg = North, 180 deg = South. **Rot az range** offers two alternatives for rotators built differently:

- **-180..+180** – the axis is centred on North and swings +/-180 deg. CardSat re-expresses each bearing into that range (e.g. 270 deg -> -90 deg), the same option Gpredict offers.
- **0.450** – the rotator has **90 deg of overlap** past North (a common Yaesu/SatPC32 arrangement). A bearing of 0-90 deg is reachable either directly or as 360-450 deg, so CardSat commands whichever is **nearer its current position**: a pass crossing North continues up into the 360-450 region instead of unwinding a full turn – no dead-zone spin at culmination.

Both affect what is sent on the **rotctld** path (e.g. **P -90.0 12.0** or **P 372.0 8.0**). GS-232 controllers natively accept 0-450 and re-wrap negatives, so the GS-232 backend follows the overlap but the +/-180 framing has no visible effect there.

For overhead passes, setting **Rot el range** to **180 deg** enables a **flip**: near culmination CardSat commands elevation past 90 deg (over the top) together with a 180 deg azimuth flip, so an antenna on a 0-180 deg elevation rotator rides through zenith without the fast 180 deg azimuth swing a 0-90 deg mount would need. Leave it at **90 deg** for a conventional elevation rotator. CardSat points **open-loop** from its own SGP4 prediction (it does not poll the controller's heading); the GS-232 **C2** read-back exists in the backend for diagnostics but is not used in the tracking loop.

Network rotators (rotctld and PstRotator)

To use the network backend, run **rotctld** on any machine that can reach your rotator (or use a controller that speaks rotctld natively), set **Rot type** to **rotctld (net)**, and enter the server's address in **Net host** (an IP is simplest) and **Net port** (Hamlib default **4533**). A dummy server for a bench test:

```
rotctld -m 1 -t 4533 -vvvvv
```

-m 1 is Hamlib's dummy rotator model and **-vvvvv** makes rotctld log every command it receives, so you can watch CardSat's **P/S** commands arrive without moving real motors (without the verbose flag the server is silent, which can look like nothing is being sent); swap in your rotator's model and serial port for the real thing. CardSat opens the socket when you enable the rotator and reconnects on its own (throttled to a few seconds) if it drops, so a server that is briefly down or rebooted recovers without intervention. Pointing is open-loop from CardSat's SGP4 prediction – it does not read the heading back.

Hardware that speaks rotctld natively works the same way, with no PC in between. For example, the **MuseLab AntRunner** (BG5DIW) portable az/el rotator (360 deg azimuth / 180 deg elevation, controlled over WiFi by its onboard ESP32) and the **AntRunner-Pro** (fixed-install, 360 deg / 90 deg, controlled over Ethernet) both present a Hamlib **rotctld**-compatible network service. Set **Rot type** to **rotctld (net)** and point **Net host** / **Net port** at the AntRunner's rotctld service (or at a **rotctld** instance bridging to it over USB).

rotctld has **no authentication** – keep it on a trusted LAN and never expose the port to the internet. Several clients may connect to one rotctld and can contend for the rotator.

PstRotator (UDP). If you already run **PstRotator** on a shack PC, CardSat can drive it directly instead of a rotctld server. In PstRotator, open **Communication > UDP Control Port**, set the port (default **12000**), and tick **UDP Control** in Setup. On CardSat set **Rot type** to **PstRotator (net)**, **Net host** to the PC's IP, and **Net port** to that UDP port – *remember to change it from rotctld's 4533 to 12000*. CardSat then sends **<PST><AZIMUTH>az</AZIMUTH><ELEVATION>el</ELEVATION></PST>** to point and **<PST><STOP>1</STOP></PST>** to stop. PstRotator does its own azimuth range, offsets and flip, so CardSat sends a plain 0-360 bearing here; leave CardSat's **Rot az range** / **Rot el range** at default and let PstRotator manage overlap and flip (or use CardSat's and keep PstRotator neutral – just don't double up).

This backend also drives the **WA4MCM PSR-100** (Don Friend, WA4MCM) – a popular portable, lightweight az/el satellite-rotor kit for field/portable work. The PSR-100 accepts the same PstRotator-style **<PST>** UDP az/el datagrams over its WiFi interface, so CardSat points it directly: set **Rot type** to **PstRotator (net)** and aim **Net host** / **Net port** at the PSR-100's UDP interface – no PC running PstRotator in between.

The PstRotator UDP control is **unauthenticated and connectionless** – keep it on a trusted LAN. Because UDP has no connection to lose, CardSat reports **Rot ON** whenever WiFi is up and a host is set; it cannot tell whether PstRotator is actually listening, so confirm the antenna follows on the first pass.

The rotator only points while you are on the Track or Polar screen. It only moves the antenna – it does not change the radio's bands, and CAT on the Yaesu/Kenwood sat rigs can't switch the band pair either, so set those by hand.

The GS-232 backend is bench-reasoned against the GS-232A/B manuals and Hamlib's gs232a/gs232b backends, and host-tested for the bridge baud math and command formatting – it has not been run against a physical SC16IS750 or a real rotator. The I2C pins (G8/G9) are confirmed from the Cap LoRa-1262 pinmap; still confirm the SC16IS750's strapped address and the controller's baud before keying real motors. The rotctld backend follows the published Hamlib `rotctld` protocol and is the one rotator path you can fully bench-test without trusting hardware encoders – point it at `rotctld -m 1` and watch the commands – but it has not driven a physical rotator either. The **PstRotator** UDP backend is host-verified for message formatting against the PstRotator manual (Rev. 7.5); it has not been run against a live PstRotator instance.

18

Managing data and factory reset

CardSat keeps all of its data in a `/CardSat` folder:

- cached GP data and per-satellite transponder caches,
- your manual GP satellites and manual transponders (kept separate, so a refresh won't erase them),
- favorites, per-satellite calibration, and all settings.

CardSat stores this on the **microSD card** by default (in the `/CardSat` folder), so your configuration travels with the card and is easy to back up. If no card is present at boot it **falls back to the internal LittleFS** flash (same `/CardSat` folder) so the unit still works standalone. The serial monitor reports which it mounted (`[fs] using microSD card for storage (/CardSat)` or `[fs] no SD card - falling back to internal LittleFS`). Insert the card **before powering on** — the filesystem is chosen once at startup. For the LittleFS fallback you must flash CardSat with a partition scheme that includes a data region, e.g. **Huge APP (3MB No OTA/1MB SPIFFS)**.

Screenshots

Press **b** on any screen to save a screenshot of exactly what's displayed. Images are written to `/CardSat/Screenshots/` on the SD card as 24-bit BMP files named `shot_0001.bmp`, `shot_0002.bmp`, ... (never overwritten). A short high beep confirms each capture; a low beep means there's no SD card to write to — screenshots require the card, as the LittleFS fallback is too small for images. The key works on every screen **except** the text-entry screen (where **b** types normally), and it is intentionally hidden — no footer hint — so it never appears in the captured image. BMPs open anywhere and convert easily to PNG for documentation.

Factory reset: Settings → **Reset all data** → type **ERASE** to confirm. On the SD card this removes CardSat's own files in `/CardSat` (it never formats your card); on internal LittleFS it formats the data partition. Either way it reboots to a clean first-run state. Use it to start over or before handing the unit to someone else.

19

Troubleshooting

No passes / “Clock not set.” Set the clock first: Update → **k** (NTP), enable GPS, or Location → **c** to enter UTC manually. Also confirm your location is set.

No transponders on Track. The satellite’s transponders weren’t cached. With WiFi, open the bird from Satellites (it fetches from SatNOGS), or use Update → **a** to cache everything. You can also add one manually (Passes → **n**).

Radio doesn’t respond / wrong VFO moves. Check the **CAT baud** (and, for Icom, the **CI-V address**) in Settings match the radio. Open the serial monitor at 115200 and watch the **CAT trace** (§16): each command is shown decoded, and the radio should reply **ACK (FB)** rather than **NAK (FA)**. If the wrong VFO tunes, the MAIN/SUB select bytes for that model may need adjusting in `radio_profiles.h`.

Radio-knob tuning feels jumpy or unresponsive. Read-back happens about twice a second, so there’s up to ~0.5 s of latency, and a 20 Hz threshold ignores tiny readback jitter. On a slower CAT link (9600 baud) it’s less snappy. If your rig quantizes frequency coarsely, that threshold may need tuning.

Predictions seem off. Check the **GP age** indicator — refresh GP data if it’s yellow/red. Confirm your location and that the clock is correct (UTC).

Fewer satellites than expected after Update. The GP file streams straight to flash and is parsed incrementally, so it isn’t limited by RAM; the catalog holds up to 220 sats. If the count looks short, open the serial monitor — the line `[net] streamed N bytes ... (declared M)` should show **N == M** (a complete download). All-null placeholder entries in the AMSAT feed (sats with no current elements) are skipped on purpose.

No alarm beeps. Confirm **AOS alarm** is on in Settings, you have **favorites**, and the **clock is set** — the alarm tracks only the soonest upcoming favorite AOS, so with no favorites or an unset clock there’s nothing to count down to.

GPS not fixing. Make sure the right **GPS source** is selected (Location → **s**) and that GPS is enabled (**p**). The Grove source shares pins with CAT — don’t use both at once.

“fs open” / “No filesystem” when downloading GP (often under a launcher). CardSat needs somewhere to store the GP file. Run directly from flash it uses the internal SPIFFS/LittleFS partition; under a launcher that didn’t attach a SPIFFS region it falls back to the **microSD card**. If you see `fs open / no filesystem`, either insert a microSD card (the launcher usually boots from one anyway), have the launcher allocate a SPIFFS partition for CardSat, or flash CardSat directly with the **Huge APP (3MB No OTA/1MB SPIFFS)** partition scheme. The serial monitor’s `[fs] ...` line at boot tells you what mounted.

Rotator shows “n/c” or won’t move. *n/c* means the I2C->UART bridge didn’t answer. Check `ROT_I2C_SDA/ROT_I2C_SCL` in `config.h` match your wiring and the SC16IS750’s address (`ROT_I2C_ADDR`) doesn’t clash with another I2C device. If it reaches the bridge but the rotator is silent, confirm the MAX3232 level shifter is in line and the controller’s baud matches **Rot baud** in Settings.



20

Screen-by-screen reference

This section catalogs **every screen** in the firmware in a uniform format — what it is for, how you get to it, what it shows, and what each key does. It complements the task-oriented walkthroughs in §8 (which follow the natural operating flow); use this section when you want the complete picture of one specific screen.

Throughout, the navigation keys are the printed arrow legends — ; up, . down, , left, / right — with **ENTER** to select and ` (or **DEL**) to go back. **b** saves a screenshot on any screen. Only the *screen-specific* keys are listed below.

Home menu



Figure 1: Home menu

- **Purpose** — the top-level launcher.
- **Reached from** — power-on lands here; ` from most top-level screens returns here.
- **Shows** — a scrolling list of the thirteen destinations: Satellites, Next Passes (all favs), Passes (sel), Track (sel), Sun / Moon, Space Wx, Weather, QRZ Lookup, Location, Update, Settings, Log, About. The header carries the clock and battery gauge.
- **Keys** — ;/. move the highlight; **ENTER** opens the selected item.

Satellites

- **Purpose** — browse the catalog, choose favorites, and reach the per-satellite analysis tools.
- **Reached from** — Home → Satellites.
- **Shows** — the satellite list (up to 220 from GP data plus any manual sats). A right-edge mark shows AMSAT activity: a filled dot = heard, filled square = telemetry only, ring = not heard, nothing = no reports. Favorites are flagged.
- **Keys** — ;/. move ({/} page); **f** toggle favorite; **v** show favorites only; **n** add a new satellite by hand; **o** open **Orbital analysis**; **s** open **Simulation**; **t** open the **Transponder database**; **d** open the **10-day pass overview**; **i** open the **Illumination** raster; **ENTER** opens **Passes** for the highlighted bird.

Satellite ID	Signal Strength	Status
*A0-07	7530	●
U0-11	14781	○
A0-27	22825	
*F0-29	24278	●
G0-32	25397	
*ISS	25544	●
N0-44	26931	●
*S0-50	27607	●
C0-55	27844	
RS-22	27939	

ENT pass o orb s sim t tx d 10d i illum

Figure 2: Satellites list

Satellite ID	Signal Strength	Status
*A0-07	7530	●
*F0-29	24278	●
*ISS	25544	●
*S0-50	27607	●
*A0-73	39444	●
*J0-97	43803	●
*RS-44	44909	●
*A0-123	61781	●
*TEVEL2-3	63218	■
*TEVEL2-7	67270	○

Favorites only

ENT pass o orb s sim t tx d 10d i illum

Figure 3: Satellites — favorites filter

Orbital analysis

NORAD	44909
Altitude	1218 km
Footprint	7325 km dia
Period	112.5 min
Apo/Peri	1511/1174 km
Fp apo/peri	8026/7209 km
Incl/Ecc	82.52 / 0.02180
SMA (a)	7721 km
B* / decay	0.000069 stable
Age/Rev	0.35d / 30188
Asc node	151.5W 15:26Z

^ bk ,// page r refresh

Figure 4: Orbital analysis — Info

- **Purpose** — a nine-page deep dive into one satellite's orbit: geometry, dynamics, lighting, and pass planning. Full theory and per-page detail are in §8 → **Orbital analysis**.
- **Reached from** — Satellites → o.
- **Shows** — one page at a time: Info, Live, Next pass, Ground track, Doppler, Nodal, Sun / Beta, Pass outlook, Orbit position.
- **Keys** — ,// flip pages; r recompute; on the **Doppler** page f sets the beacon frequency; ` leaves.



Figure 5: Orbital analysis — Live



Figure 6: Orbital analysis — Next pass



Figure 7: Orbital analysis — Doppler



Figure 8: Orbital analysis — Nodal

```

RS-44 Sun/B 14:33:48Z
Beta now      -57.9 deg
Sunlight      full-sun orbit
Beta*         +/-34.3 deg (full-sun)
Eclipse       none (continuous sun)
-> eclipses    07/22 (39d)

^ bk ,// page r refresh

```

Figure 9: Orbital analysis — Sun/Beta

```

RS-44 Outloo 14:33:52Z
Next 7 days
Passes        46 (19 >30 deg)
Longest       22.2 min
Avg gap       3.6 h
Best el       82 deg
Best at       06/15 10:36Z
Best in       2643:03
Best dur      22:15

^ bk ,// page r refresh

```

Figure 10: Orbital analysis — Pass outlook

```

RS-44 OrbPos 14:33:54Z
Mean anom     40.6 deg
True anom     42.3 deg
Arg lat       195.6 deg
To perigee    99:49
To apogee     43:33
Arg peri      153.3 deg
RAAN          342.3 deg
Rev now       30188
Epoch age    0.35 d

^ bk ,// page

```

Figure 11: Orbital analysis — Orbit position

```

RS-44 sim 14:34:02Z
2026-06-13 14:34:02Z
Offset        +0d 00h 00m
Step          1 hr
Az / El       82.6 / -58.9
Range         12309 km
Sub pt        -15.30,43.70
Altitude      1223 km
below horizon sunlit

^ bk m map ,// time ;//. step x now

```

Figure 12: Simulation



Figure 13: Simulation — map view

Simulation

- **Purpose** — a “time machine” that propagates the selected satellite to an arbitrary time so you can preview geometry past or future.
- **Reached from** — Satellites → **s**.
- **Shows** — the sub-point and footprint (and, in world-map view, the track) at the simulated instant, with the offset from now.
- **Keys** — **,** **//** step the time backward/forward; **;/.** change the step size; **m** toggle the world-map view (sub-point + footprint); **x** reset to now; **`** back.

Transponder database

- **Purpose** — inspect every transponder/beacon entry CardSat holds for a satellite (from SatNOGS plus any you added).
- **Reached from** — Satellites → **t**.
- **Shows** — a scrollable list of entries with description, downlink (and mode), uplink, and tone/inverting flags. The currently selected entry is highlighted with a **>**; entries you added by hand are tagged with a *****.
- **Keys** — **;/.** select an entry; **x** **delete** the selected entry, *only if it's a manual one* (*-tagged) — press **x** once to arm, **x** again to confirm. **`** back. Deleting rewrites that satellite's `/CardSat/mtx_<norad>.json` file; SatNOGS-cached entries can't be deleted here (they'd return on the next Freq update). To **edit** a manual transponder, delete it and re-add it with **n** on the **Passes** screen.

EQX table (OSCARLOCATOR)

- **Purpose** — a table of **equatorial crossing** (EQX) times and longitudes for the selected satellite, for use with a classic **OSCARLOCATOR** plotting board. Each EQX is an **ascending-node** crossing — the moment the satellite's ground track crosses the equator heading **north**. With the EQX time and longitude you rotate the Oscarlocator's track overlay to that longitude and read AOS/LOS and mutual visibility directly off the board, with no live computer at the operating position. A **d** keypress switches the table to the **descending node** (southbound equator crossing) if you'd rather reference that.
- **Reached from** — Satellites → **e**.
- **Shows** — a scrollable, day-grouped table covering the next **3 days**, each row an EQX **date, time in UTC**, and **sub-satellite longitude in West-positive notation** (**123.4 W**),

matching the convention printed on Oscarlocator dials. Successive crossings step westward by roughly $360^\circ \times (\text{period}/\text{sidereal-day})$ per orbit ($\sim 28.7^\circ$ for an AO-7-class orbit).

- **Computed on-device** from the satellite's current GP elements (SGP4) — no network needed — so the longitudes drift as the elements age; run **Update** every week or two to keep them fresh, the same as for tracking.
- **Keys** — `;/.` scroll a page at a time; `d` **toggle ascending ↔ descending** node (recomputes the table — ascending/EQX is the default and the usual OSCARLOCATOR reference, descending gives the southbound equator crossing); `r` recompute (e.g. after the clock is set or new elements are loaded); ``` back. Requires the **UTC clock** to be set (GPS or Location → `c`). The header shows **EQX** for ascending or **DEQX** for descending.

The table mirrors the output of the **AO-7_OSCARLOCATOR** generator (github.com/prstoetzer/AO-7_OSCARLOCATOR), but works for any satellite in the catalog and runs entirely on the Cardputer.

Next Passes (schedule)

When	Satellite	El	Len
NOW	JO-97	0	0m
9m	TEVEL2-7	28	10m
41m	RS95S (QMR-KW)	19	10m
56m	AO-07	4	10m
1h15	TEVEL2-3	34	10m
1h19	ISS	12	9m
5h03	AO-73	26	11m
5h19	RS-44	10	14m
6h04	SO-50	6	8m

ENT trk m map r refr z slp ` bk

Figure 14: Next Passes schedule

- **Purpose** — the unified upcoming-pass schedule across **all** your favorites, so you see what is next regardless of which bird it is.
- **Reached from** — Home → Next Passes (all favs).
- **Shows** — favorites' passes in time order (up to 24 favorites tracked), each with satellite, AOS time/countdown, max elevation and duration. Stale-element sats carry a red !.
- **Keys** — `;/.` select; **ENTER** opens **Pass detail**; `m` opens the live **World map**; `z` arms **deep sleep** until the next AOS; ``` back.

Passes

- **Purpose** — the pass list for one selected satellite, and the jumping-off point to tracking, the visualizers, and the workable lists.
- **Reached from** — Satellites → **ENTER**, or Home → **Passes (sel)**.
- **Shows** — up to twelve upcoming passes for the bird (AOS/LOS, max elevation, duration).
- **Keys** — `;/.` select; `d` **Pass detail**; `t` or **ENTER** open **Track**; `n` add a manual transponder; `r` recompute; `x` **Mutual windows**; `v` **10-day overview**; `i` **Illumination**; `g/w/e` workable grids / US states / DXCC for this pass; ``` back.

Pass detail

- **Purpose** — the full numeric breakdown of one pass plus its polar plot.



AOS (UTC)	Dur	El	LOS	GP
06/13 19:45	14m	10	20:00	1.7d
06/13 21:37	19m	72	21:57	
06/13 23:32	16m	17	23:48	
06/14 08:25	14m	10	08:40	
06/14 10:13	22m	67	10:35	
06/14 12:08	20m	25	12:28	
06/14 20:08	17m	19	20:25	
06/14 22:01	19m	71	22:21	
06/14 23:57	12m	9	00:09	

ENT trk d dtl n+TX g grd w st e dx x mut

Figure 15: Passes (per-satellite)

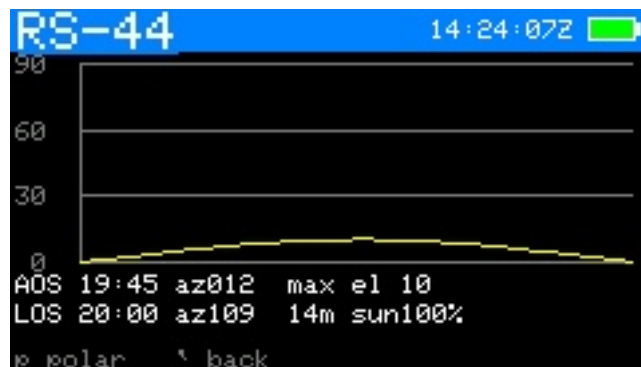


Figure 16: Pass detail (elevation)

- **Reached from** — Passes → d, or Next Passes → ENTER.
- **Shows** — AOS/TCA/LOS times and azimuths, max elevation, duration, and a polar (sky) plot of the arc.
- **Keys** — p toggle to the dedicated **Pass polar** view; ` back.

Pass polar



Figure 17: Pass polar plot

- **Purpose** — a full-screen polar (azimuth/elevation sky) plot of a single pass's arc.
- **Reached from** — Pass detail → p.
- **Shows** — the pass traced on a polar grid (zenith centre, horizon rim), with AOS/LOS marked.
- **Keys** — p toggle back to Pass detail; ` back.

Mutual windows (co-visibility)



Figure 18: Mutual windows

- **Purpose** — find the times a satellite is simultaneously visible to you **and** to a second station — the windows in which a contact is geometrically possible.
- **Reached from** — Passes → x.
- **Shows** — the other station's grid (editable) and a list of mutual-visibility windows with start/end and the overlap duration.
- **Keys** — ;/. scroll the windows; **ENTER** edit the remote grid; ` back.

10-day pass overview



Figure 19: 10-day pass overview

- **Purpose** — an at-a-glance visibility chart of one satellite's passes over ten days, modelled on InstantTrack's multi-day view.
- **Reached from** — Passes → v (or Satellites → d).
- **Shows** — one row per day (today at top), each a 24-hour UTC timeline; every pass is a bar shaded by peak elevation (dim-green < 15°, green < 40°, yellow above). A red tick marks now.
- **Keys** — ;/. scroll one day (forward indefinitely, not before today); r recompute; ` back to Passes.

Illumination

- **Purpose** — a 60-day solar-illumination raster (DK3WN *illum* style) showing when the satellite is sunlit vs. eclipsed across its orbit.
- **Reached from** — Passes → i (or Satellites → i).



Figure 20: Illumination

- **Shows** — horizontal axis date (today → +60 d), vertical axis one orbital period; cells yellow in sunlight, dark in eclipse. A live readout shows current SUN/SHADOW status, eclipse minutes/percent per orbit, and time to the next transition.
- **Keys** — ,// scroll one day through the window (forward indefinitely, not before today); r recompute; ` back to Passes.

Track



Figure 21: Track — live Doppler & CAT



Figure 22: Track — ground track

- **Purpose** — the main operating screen: live pointing, Doppler-corrected frequencies, transponder selection, calibration, and radio/rotator control. Full detail in §8 → Track, Doppler theory in §9.
- **Reached from** — Passes → t/ENTER, or Home → Track (sel).

- **Shows** — az/el (and GP age), range/range-rate (ECL flag in eclipse), the transponder, DN/RX and UP/TX frequencies, the passband line, the calibration line, the radio status, a rotator status line, and a **PL** tone line on FM birds that need one.
- **Keys** — **m** switch TUNE/CAL; **d** cycle Doppler tune mode (linear birds); **t** next transponder; **c** set CTCSS/PL tone; **r** radio output on/off; **o** rotator on/off; **p** **Polar**; **f** **Manual mode**; **l** **Log QSO**; **g/w/e** workable grids / US states / DXCC now; in TUNE: **,** **/** move spot, **s** step, **x** recenter; in CAL: **,** **/** trim downlink, **;** **/** trim uplink, **s** step, **x** zero; **ENTER** save calibration.

Manual mode

- **Purpose** — a no-radio frequency calculator: CardSat shows the Doppler-corrected dial frequencies so you can tune a non-CAT rig by hand.
- **Reached from** — Track → **f**.
- **Shows** — the same RX/TX figures as Track, formatted for manual dialing, with the passband/tune controls but no CAT output.
- **Keys** — mirror Track's tuning set: **m**, **t**, **s**, **x**, the **,** **/**, **;** **/** tune/cal keys, **g/w/e** workable lists, **l** log, **p** polar, **u/e/f** as on Track; **`** back.

Polar

- **Purpose** — a live full-screen polar sky plot of the satellite you are tracking, for visual aiming.
- **Reached from** — Track → **p**.
- **Shows** — the current az/el as a marker on a polar grid, with the pass arc.
- **Keys** — **p** toggle back to Track; **l** **Log QSO**; **`** back.

Workable grids

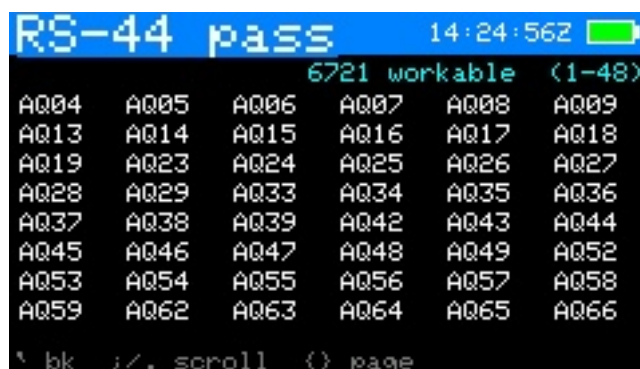


Figure 23: Workable grids

- **Purpose** — the Maidenhead grid squares currently (or, from Passes, during the pass) reachable through the satellite's footprint.
- **Reached from** — Track → **g** (live) or Passes → **g** (for the pass).
- **Shows** — the list/scatter of workable grids; from Track it refreshes live while radio and rotator control keep running.
- **Keys** — **;** **/** scroll; **`** back.



Figure 24: Workable US states

Workable US states

- **Purpose** — the US states (for WAS chasing) under the footprint now or during the pass.
- **Reached from** — Track → w (live) or Passes → w (for the pass).
- **Shows** — the workable states list.
- **Keys** — ;/. scroll; ^ back.

Workable DXCC



Figure 25: Workable DXCC

- **Purpose** — the DXCC entities under the footprint now or during the pass.
- **Reached from** — Track → e (live) or Passes → e (for the pass).
- **Shows** — the workable DXCC list (derived from bundled cty.dat geometry).
- **Keys** — ;/. scroll; ^ back.

Sun / Moon

- **Purpose** — point the rotator at the Sun or Moon for sun-noise/EME work and antenna calibration; full detail in §13 → Sun / Moon.
- **Reached from** — Home → Sun / Moon.
- **Shows** — the selected body's live az/el (and a sky-dome graphic in graphic view). An orange SUN/MOON header tag appears while it is driving the rotator.
- **Keys** — ;/. switch Sun↔Moon; g toggle graphic/list view; o rotator tracking on/off; x stop; ^ park and back.



Figure 26: Sun / Moon

Space weather



Figure 27: Space weather

- **Purpose** — solar flux and geomagnetic indices with a plain-language operating outlook; detail in §13 → [Space weather](#).
- **Reached from** — Home → Space Wx.
- **Shows** — F10.7 flux, planetary Kp, running A index, each labelled and colour-coded, an HF/satellite outlook line, and a data-freshness note.
- **Keys** — r refresh over WiFi; ` back.

Weather



Figure 28: Weather

- **Purpose** — terrestrial current conditions and a multi-day forecast for the operating site; detail in §13 → [Weather](#).

- **Reached from** — Home → Weather.
- **Shows** — current temperature, sky, wind and humidity, then per-day condition, high/low and precipitation chance, plus a freshness note. Units per Settings.
- **Keys** — `r` refresh over WiFi; ``` back.

QRZ callsign lookup



Figure 29: QRZ callsign lookup

- **Purpose** — resolve a callsign to name, location, grid and licence class via the QRZ.com XML API; detail in §13 → QRZ callsign lookup.
- **Reached from** — Home → QRZ Lookup.
- **Shows** — prompts for credentials/WiFi if needed; otherwise the looked-up station's details.
- **Keys** — `ENTER` type a callsign to look up; `;/.` scroll a long result; ``` back.

Location

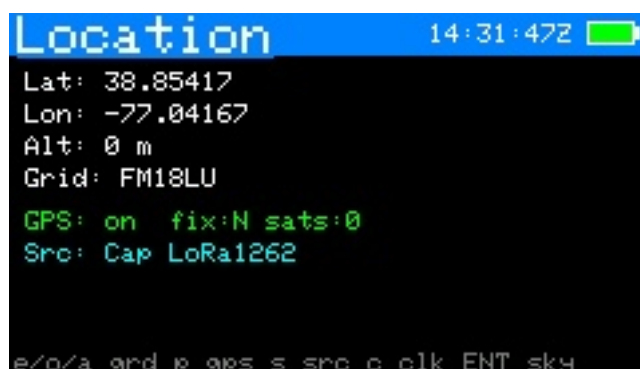


Figure 30: Location

- **Purpose** — set your station position and clock, and view the GPS sky plot.
- **Reached from** — Home → Location.
- **Shows** — current lat/lon/altitude, grid, time source and clock, and GPS status when enabled.
- **Keys** — `e` edit latitude; `o` edit longitude; `a` edit altitude; `g` enter a Maidenhead grid; `p` enable/disable GPS; `s` pick the GPS source; `c` set the UTC clock by hand; `ENTER` open the GPS sky plot; ``` back.



Figure 31: GPS sky plot

GPS sky plot

- **Purpose** — a polar plot of the GNSS satellites currently in view, by signal strength — useful for checking antenna/fix quality.
- **Reached from** — Location → ENTER.
- **Shows** — GNSS satellites placed by azimuth/elevation, coloured green (strong) to grey (weak), with fix data.
- **Keys** — ` back. (The plot updates live; no other keys.)

Update

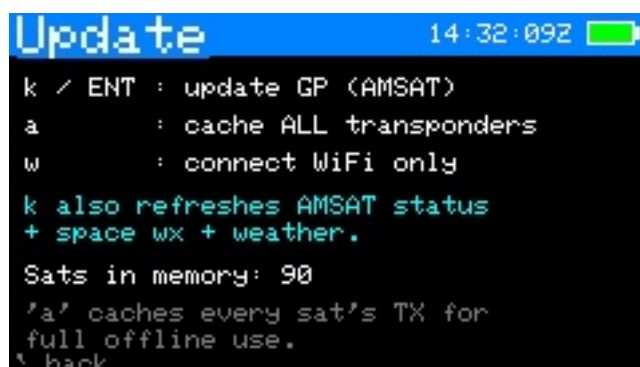


Figure 32: Update

- **Purpose** — refresh everything that comes from the network in one place.
- **Reached from** — Home → Update.
- **Shows** — the last GP age and a note that k also refreshes the clock, AMSAT status, space weather and terrestrial weather.
- **Keys** — k (or ENTER) download GP + sync clock (NTP) + AMSAT + space wx + weather; a fetch and cache **all** transponders for offline use; ` back.

GP source

- **Purpose** — choose where element sets come from.
- **Reached from** — Settings → GP source → ENTER.
- **Shows** — a picker: AMSAT (amateur, default), the CelesTrak category groups (Amateur first, then SatNOGS and the other groups), and Custom URL.
- **Keys** — ;/. move ({/} page); ENTER select; ` back. The choice is saved immediately and used by the next Update.

Settings



Figure 33: Settings menu



Figure 34: Settings — Radio / CAT

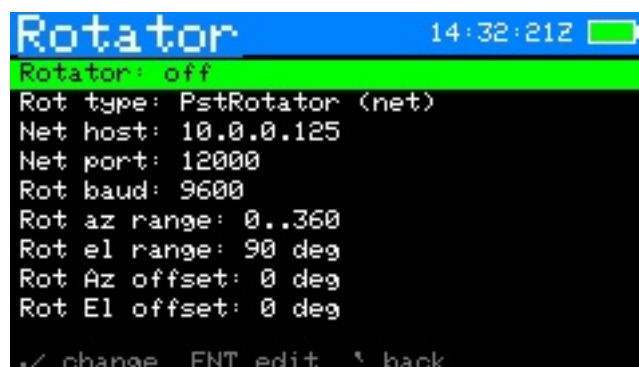


Figure 35: Settings — Rotator

- **Purpose** — all configuration, grouped into four submenus (Radio / CAT, Rotator, Station / display, Network). Each row and its adjust keys are tabulated in §8 → Settings.
- **Reached from** — Home → Settings.
- **Shows** — the submenu list, then the rows within a chosen submenu with their current values.
- **Keys** — ; / . move; ENTER open a submenu / edit a text field; , // change a value in place; ` back.

WiFi scan

- **Purpose** — pick a network from a live scan instead of typing the SSID.



Figure 36: Settings — Station / display



Figure 37: Settings — Network / data

- **Reached from** — Settings → WiFi SSID row → s.
- **Shows** — nearby networks, strongest first, * marking secured ones.
- **Keys** — ;/. select; **ENTER** choose (then enter the password unless the network is open); r rescan; ` back.

Rotator manual / calibration

- **Purpose** — jog the rotator by hand with live read-back, and — for a **Yaesu (direct)** rotator — calibrate the position ADC. Detail in §17 and ROTOR_INTERFACE.md.
- **Reached from** — Settings → Rotator: manual control → ENTER.
- **Shows** — commanded and actual az/el; for Yaesu direct, the live ADC counts for each axis.
- **Keys** — ,// jog azimuth; ;/. jog elevation; s step size; x stop; (Yaesu direct only) 1/2/3/4 capture the ADC counts at az 0 / az full / el 0 / el full; ` back.

Log menu

- **Purpose** — the QSO logging hub.
- **Reached from** — Home → Log.
- **Shows** — New QSO entry, View / edit log, Export to ADIF.
- **Keys** — ;/. move; **ENTER** open the selected item; ` back.

Log entry

- **Purpose** — create or edit one QSO record. Full field list in §8 → Logging QSOs.

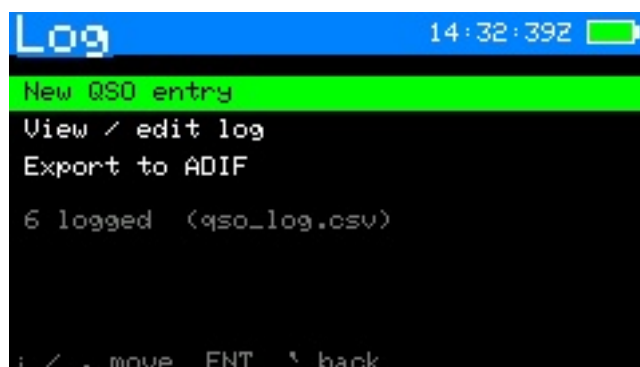


Figure 38: Log menu



Figure 39: Log entry

- **Reached from** — Track/Polar/Manual → **l**, or Log menu → New QSO entry, or Log list → ENTER on a record.
- **Shows** — every editable field: date, time, sat, mode, DL/UL MHz, call, RST sent/received, grid, notes.
- **Keys** — **;/.** move between fields; **ENTER** edit the selected field (the Sat field opens the satellite picker; Mode toggles SSB/CW on linear birds); **s** save; **^** cancel.

Log list (view / edit)



Figure 40: Log list

- **Purpose** — review and correct stored contacts.
- **Reached from** — Log menu → View / edit log.
- **Shows** — a scrollable list of the most recent 120 QSOs.

- **Keys** — `;/.` move; **ENTER** open a record to edit; (within a record) `x` twice to delete; ``` back.

World map

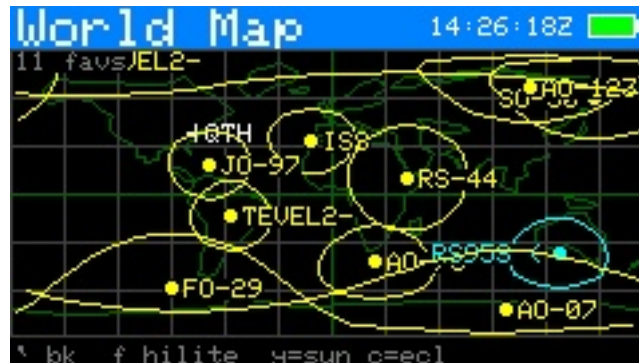


Figure 41: world-map

- **Purpose** — a live equirectangular map of all favorites' footprints with the day/night terminator.
- **Reached from** — Next Passes $\rightarrow$ `m`.
- **Shows** — every favorite's sub-point and footprint, your station marker, the graticule, and the sun terminator (drawn automatically).
- **Keys** — `f` cycle which favorite is highlighted; ``` back.

Help

- **Purpose** — an on-device scrollable key reference, available almost anywhere.
- **Reached from** — `h` on most screens.
- **Shows** — per-screen key summaries.
- **Keys** — `;/.` scroll; ``` back.

About

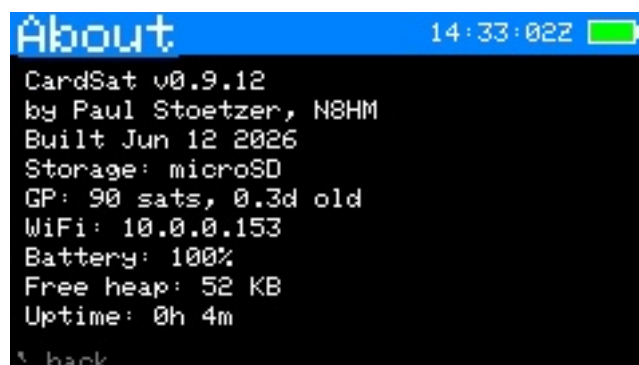


Figure 42: About

- **Purpose** — build and diagnostic information.
- **Reached from** — Home $\rightarrow$ About.
- **Shows** — firmware version, IP address, free heap and other read-only diagnostics.
- **Keys** — ``` back.

Edit (text/number entry)

- **Purpose** — the shared on-screen editor used by every field that needs typed input (frequencies, SSID/password, grid, callsign, host/port, clock, etc.).
 - **Reached from** — automatically, whenever a field is opened for editing.
 - **Shows** — the field title, the current buffer, and the entry footer. Certain fields default to **uppercase** (callsign, grid, SSID and similar) — hold shift for lowercase.
 - **Keys** — type to enter; **DEL** backspace; **ENTER** accept; **`** cancel.
-

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Key reference (cheat sheet)

Global: ; up · . down · , left · / right · ENTER select · ` /DEL back · { / } page · **b** screenshot · **h** help.

Screen	Keys
Satellites	f favorite · v favorites-only · n new GP sat · x delete manual sat (2-press) · e EQX table (OSCARLOCATOR) · o orbital analysis · s simulation · t transponder database · d 10-day overview · i illumination · ENTER passes · right-edge AMSAT mark: filled dot = heard, filled square = telemetry only, ring = not heard, none = no reports
Orbital analysis	, // page (Info / Live / Next pass / Ground track / Doppler / Nodal / Sun-Beta / Pass outlook / Orbit position) · Info: footprint diameter now/apogee/perigee (= longest possible QSO) + decay estimate & solar-bracket range · Live: az/el/range/Doppler, mean anomaly/phase, sunlit/eclipse + eclipse depth (deg; >0 = in shadow) · Next pass: slant ranges + path delay + peak eclipse depth · Doppler: f set beacon freq, peak shift + max range-rate · Nodal: J2 node/perigee drift, sun-sync, LTAN, repeat track, longest pass · Sun/Beta: solar beta angle, full-sun vs eclipsed, eclipse %/orbit, next transition · Pass outlook: 7-day pass count/>30° count/longest/avg gap + the best upcoming pass (elevation, when, duration) · Orbit position: mean/true anomaly, argument of latitude, time to perigee/apogee, RAAN, rev number · r recompute · ` back
Simulation	, // step time · ; / step size · m world-map view (sub-point + footprint at the simulated time) · x reset to now · ` back
Next Passes	ENTER track · m world map · r refresh · z deep-sleep until AOS

Screen	Keys
Passes	;/. select · d detail · t /ENTER track · n add TX · r recompute · x mutual · v 10-day · i illum · g workable grids (this pass) · w workable US states (this pass) · e workable DXCC (this pass)
Pass detail	p polar of this pass · ` /ENTER back
Pass polar	p back to curve · ` /ENTER passes
Track	m TUNE/CAL · d cycle tune mode (FULL/DL/UL/hold) · t next TX · c CTCSS tone · r radio on/off · o rotator on/off · p polar · f Manual mode · l log QSO · g workable grids now · w workable US states now · e workable DXCC now (radio/rotator keep running) · ENTER save cal
Manual mode	no-radio frequency calculator · u toggle which leg is fixed (HOLD vs TUNE>) · , / // move fixed freq in passband (linear) · s step · x center · m CAL · t next TX · l log · p polar · g grids (all return here) · ENTER save cal · ` / f back to Track
Workable grids	4-char Maidenhead grids under the footprint (per-pass union or live, refreshing ~3 s; uncapped, works to high orbits) · count shown on a cyan line above the list · ;/. and {/} scroll · ` back
Track · TUNE	, / // tune $\mp$ · s step (100/1k/5k) · x recenter
Track · CAL	, / // downlink $\mp$ · ;/. uplink $\mp$ · s step (10/100/1k) · x zero
Polar	l log QSO · p /ENTER/ ` back to track
Log (menu)	;/. select · ENTER → new QSO / browse / export ADIF
Log · list	;/. scroll · ENTER edit entry · ` back
Log · entry	;/. field · ENTER edit · s save · x ×2 delete · ` back
Mutual	;/. scroll · ` /ENTER back to passes
10-day	;/. scroll $\mp$ 1 day (forward indefinitely, oldest day off the top; not before today) · r recompute · ` /ENTER back

Screen	Keys
Illum	,// scroll $\mp$ 1 day (forward indefinitely; not before today) · r recompute · ` /ENTER back
Location	e/o/a lat/lon/alt · g grid · p GPS on/off · s GPS source · c set clock · ENTER GPS sky plot
GPS sky plot	live GNSS az/el coloured by signal · ` back
World map	f cycle highlighted favorite (sun terminator drawn automatically) · ` back
Rotator (manual)	,// az · ;/. el · s step · x stop · (<i>Yaesu direct only</i>) 1/2/3/4 capture ADC at az 0 / az full / el 0 / el full · ` back
Help	;/. scroll · ` back
Update	k /ENTER GP · a cache all TX · w WiFi only
Settings	,// change · ENTER edit/toggle · s scan WiFi (on SSID row) · (Reset = type ERASE)
GP source	pick AMSAT / any CelesTrak JSON-PP category (Amateur Radio first) / Custom URL · ;/. move · {/} page · ENTER select
Sun / Moon	graphical sky-dome view (Sun/Moon glyphs on a polar dome) · g toggle graphic/data list · ;/. pick Sun/Moon · o rotor track on/off (takes the rotator from sat tracking) · auto-parks while the body is below the horizon · header shows SUN/MOON tag on other screens · x stop · ` back
Space Wx (main menu)	solar 10.7 cm flux + planetary Kp + running A index, each labelled & colour-coded, with a plain-language HF/satellite operating outlook and a data-freshness note · r refresh over WiFi · ` back
Weather (main menu)	terrestrial current conditions + multi-day forecast for the operating site from Open-Meteo · current temp/sky/wind/humidity then per-day hi/lo & precip chance · refreshes on entry (if on WiFi) and with Update · r refresh · cached offline · ` back

Screen	Keys
QRZ Lookup (main menu)	callsign lookup via QRZ.com XML (needs a QRZ XML subscription + credentials in Settings → Network) · ENTER type a callsign · shows name/address/country/grid/class · WiFi required · ` back
Transponder DB (Satellites → t)	scrollable list of the selected satellite's transponder/beacon entries (description; D downlink + mode; U uplink + tone/inv/lin flags) · ;/. scroll · ` back
Edit	type · DEL backspace · ENTER ok · ` cancel
About	build/version, IP, free heap and diagnostics (read-only)

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Glossary

- **AOS / LOS** — Acquisition / Loss of Signal: when the satellite rises above and sets below your horizon.
 - **TCA** — Time of Closest Approach (maximum elevation).
 - **Azimuth / Elevation** — compass bearing (0° =N, clockwise) and angle above the horizon to the satellite.
 - **Range / range-rate** — slant distance to the satellite and how fast it's changing (drives Doppler; +ve = receding).
 - **Doppler shift** — the frequency change caused by relative motion.
 - **Downlink / uplink** — the satellite's transmit frequency (you receive) and receive frequency (you transmit).
 - **Linear transponder** — relays a band of frequencies (SSB/CW), as opposed to a single FM channel. May be **inverting** (flips the spectrum) or non-inverting.
 - **Passband** — the range of frequencies a linear transponder relays.
 - **GP / OMM** — General Perturbations data / Orbit Mean-elements Message: the orbital element set (distributed as JSON) that SGP4 propagates. Replaces the legacy **TLE** (Two-Line Element) text encoding, which CardSat still rebuilds internally to feed the propagator.
 - **SGP4** — the standard model that turns an element set into position/velocity over time.
 - **Maidenhead grid** — the locator system (e.g. **FM18**) for your station position.
 - **Eclipse / sunlit** — whether the satellite is in Earth's shadow or illuminated.
 - **CAT** — Computer Aided Transceiver control: the generic term for computer control of a radio. CardSat speaks three CAT dialects — Icom **CI-V**, Yaesu, and Kenwood — behind one abstract rig interface, over a wired serial bus or (for the **IC-9700**) the **Icom LAN RS-BA1** protocol.
 - **CI-V** — Icom's Communications Interface-V serial control bus (Icom's CAT dialect).
 - **MAIN / SUB** — the two VFOs/bands of a satellite-capable radio; by default CardSat uses MAIN for uplink and SUB for downlink (the **VFO Type** setting swaps them).
 - **One True Rule** — KB5MU's principle: tune so the frequency *at the satellite* stays constant; the computer corrects both legs for Doppler.
-

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Supporting AMSAT

CardSat exists because of the work AMSAT and its volunteers do — building and keeping amateur satellites flying, publishing the orbital data this app depends on, and advocating for amateur radio in space. **If you find CardSat useful, please consider joining and/or donating to AMSAT at www.amsat.org.** Membership and donations directly fund the next generation of satellites you'll track and work with this very tool.

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